

Digitalization, Factor-Biased Productivity, and Market Power*

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Abstract

Digitalization may improve labor efficiency, raise productivity across all inputs, or allow firms to charge higher prices relative to marginal costs. This paper develops a novel method that simultaneously estimates labor-augmenting productivity, Hicks-neutral productivity, and markups. Our framework allows each component to be persistent overtime and allow digitalization to be endogenous. Under perfect competition, we identify the technology parameters using a proxy-variable identification approach. We then extend this approach to firms with market power and identify the technology parameters without imposing a particular model of firm competition. The framework also separates transitory output noise from persistent productivity. We apply the model to Spanish manufacturing firms over 2000–2019 using a multidimensional index of digitalization. Under perfect competition, a 0.10-point increase in digitalization is associated with 0.37% higher labor-augmenting productivity and 0.62% higher Hicks-neutral productivity in the following period. Under imperfect competition, the corresponding effects are 0.55% and 0.43%, alongside a 0.07% increase in markups. The labor-augmenting response is significantly larger for small firms under both specifications, while the markup response is significantly smaller in highly digitalized industries. These results show that the effects of digitalization cannot be summarized by a single productivity measure. Distinguishing labor-augmenting productivity, Hicks-neutral productivity, and markups reveals whether digitalization improves production efficiency or firms' ability to price above marginal cost.

Keywords: Digitalization, Factor-Biased Productivity, Labor-Augmenting Productivity, Markups, Market Power, Spanish Manufacturing

JEL Classification: D24, L13, L60, O33, J23, C33

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1 Introduction

Digital technologies can improve the efficiency of particular inputs, raise overall productive efficiency, or change firms’ ability to price above marginal cost. These channels have different implications for firms, workers, and consumers. Distinguishing them is therefore essential to understanding the economic consequences of digitalization. Industry surveys indicate that managers increasingly view digital adoption as a means of improving operating efficiency and financial performance (PwC, 2026; Gartner, 2026).

This paper studies how digitalization is associated with the evolution of labor-augmenting productivity, Hicks-neutral productivity, and firm markups. We use panel data on Spanish manufacturing firms from the Encuesta Sobre Estrategias Empresariales (ESEE) covering 2000–2019. We compare a perfect-competition benchmark with a specification that allows firms to have market power. To examine differences across technologies, we construct a 13-variable digitalization index and four component indices: IT technologies, digital human capital, automation, and stakeholder-facing technologies.

Our findings show positive relationships between digitalization and both productivity components, alongside a small markup response. Under imperfect competition, a 0.10-point increase in the digitalization index is associated with approximately 0.55% higher labor-augmenting productivity, 0.43% higher Hicks-neutral productivity, and 0.07% higher markups in the following period, conditional on their previous levels. The overall productivity relationships remain positive and significant under perfect competition. Under both specifications, the labor-augmenting response is significantly larger among small firms than among large firms. Higher adoption levels among large firms therefore need not imply larger incremental labor-efficiency gains.

These results extend the positive digitalization–productivity relationship documented by Gal et al. (2019) by distinguishing labor-augmenting improvements from Hicks-neutral gains. They also qualify the interpretation of the higher markups in digitally intensive industries reported by Calligaris et al. (2018). In our estimates, additional firm digitalization has a significantly smaller markup response in high-digital-intensity industries than in low-digital-intensity industries. The studies examine different objects, higher industry markup levels need not imply a larger response to further adoption. Our empirical contribution is to distinguish these responses and show how they vary across firms and technologies.

Identifying these relationships requires separating technology from market power. Much of the digitalization literature measures total factor productivity while treating technological change as Hicks-neutral. This can affect improvements in the efficiency of particular inputs. Omitting factor-biased productivity can also distort output elasticities and production-based markup estimates (Doraszelski and Jaumandreu, 2019; Demirer, 2026). This matters for approaches such as De Loecker and Warzynski (2012), which recover markups using output elasticities and revenue expenditure shares. Incorrectly specified elasticities can lead researchers to attribute technological changes to changes in market power.

Our methodological contribution is a sequential estimator that identifies technology before recovering markups. We first identify the relevant restricted translog technology parameters from the quantity side. We then use the revenue-share equation to construct a firm-year markup signal. Under additional second-

moment restrictions, a state-space model uses Kalman filtering and smoothing to separate transitory noise from persistent Hicks-neutral productivity and markups. This approach combines structural identification with signal extraction without requiring firm-level input-price data or specifying the form of strategic interaction among firms.

The findings also inform the evaluation of digital adoption policies. The stronger labor-augmenting response among small firms motivates examining support that improves their access to digital technologies and relevant skills. The component estimates identify IT technologies, digital human capital, and automation as relevant areas for such evaluations. Potential benefits include greater labor efficiency and productive capacity. Establishing whether these gains reach workers and consumers requires evidence on wages, employment, prices, and adoption costs alongside productivity and markups. Our estimates provide quantitative benchmarks for these assessments, while causal returns to specific interventions require additional evidence.

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and digitalization measures. Section 4 presents the model and identification strategy. Section 5 reports the main results and robustness analysis. The appendices provide the alternative materials-demand specification and detailed variable definitions.

2 Related Literature

The paper connects four related literatures: digitalization and productivity, production function estimation, factor-biased technology, and firm-level markups. The main empirical literature on digitalization generally studies total factor productivity, labor productivity, or other aggregate outcomes. A large group of studies reports positive productivity effects of ICT and digital adoption (Nucci et al., 2023; Cardona et al., 2013; Brynjolfsson and Yang, 1996; Brynjolfsson and Hitt, 2000; Baily, 2002; Dedrick et al., 2003; Melville et al., 2004; Stiroh, 2005; Draca et al., 2008; Holt and Jamison, 2009). Other studies find weak or insignificant effects (Acemoglu et al., 2014; DeStefano et al., 2018; Haller and Lyons, 2015; Bertschek et al., 2015; Colombo et al., 2013). For example, Acemoglu et al. (2014) find little evidence that digital intensity raises labor productivity outside the computer-producing sector of U.S. manufacturing.

The empirical methods in this literature also differ. Nucci et al. (2023) use difference-in-differences and propensity-score matching. Gal et al. (2019) use a control-function approach. Anderton et al. (2023) study a panel of EU countries, while Cusolito (2020) use firm-level data from developing countries. These studies are informative about the overall productivity effect of digitalization. They generally do not separate labor-augmenting productivity from Hicks-neutral productivity.

That distinction is central for us. Production-function methods that recover firm-level markups also depend on reliable output elasticities. If technology changes along more than one dimension, an aggregate productivity residual may mix distinct technological channels. The same problem can affect markup measurement because revenue shares reflect both technology and market power. Our framework therefore treats factor-biased productivity and markups jointly rather than interpreting either from a single

residual.

This paper also contributes to the literature on production-based markup estimation. A widely used approach is developed by De Loecker and Warzynski (2012), who recover markups from firms’ cost-minimization conditions using production data and at least one freely variable input. Their approach relates the markup to the ratio of an input’s output elasticity to its revenue expenditure share, with an adjustment for unanticipated output shocks, without specifying the form of competition in the product market. The reliability of the resulting markup estimates nevertheless depends on the identification and specification of output elasticities. Doraszelski and Jaumandreu (2019) highlight two difficulties: unobserved markups can complicate productivity-proxy inversion under IPC, and conventional production-function specifications may fail to reconcile observed input expenditure shares with the implied output elasticities. Similarly, Demirer (2026) shows that neglecting labor-augmenting productivity can distort estimated output elasticities and, consequently, markups. Relatedly, Raval (2019) finds that markups estimated using materials are negatively correlated with those estimated using labor and shows that non-neutral productivity differences across firms can explain this discrepancy. Flynn et al. (2019) show that production data alone do not identify markups without additional restrictions.

To address these concerns, Doraszelski and Jaumandreu (2019) combine cost-minimization conditions with a restricted translog technology that accommodates both labor-augmenting and Hicks-neutral productivity. They consider a two-stage procedure that uses the lagged inverted production function to control for Hicks-neutral productivity, as well as a simultaneous estimator of technology and markups. Our IPC model separates factor-biased productivity from market power using a distinct identification strategy. We first identify the technology block from the quantity side and then use the revenue-share equation to obtain a firm-year markup signal. A state-space representation with Kalman filtering and smoothing then separates transitory noise from the persistent states and recovers observation-specific estimates of markup and Hicks-neutral productivity. Section 5.5 provides the full identification argument and comparison with the existing literature.

Other studies address markup identification through restrictions on returns to scale or alternative estimation frameworks. Flynn et al. (2019) show that prior knowledge of returns to scale can resolve the identification problem in production-based markup estimation. When external information on returns to scale is unavailable, they propose imposing constant returns as a practical identifying restriction. Using a different approach, Kumbhakar et al. (2012) develop a stochastic-frontier framework for estimating market power without requiring input-price data. Their framework accommodates estimation both with and without imposing constant returns to scale. Together, these studies highlight the importance of specifying the technological and statistical restrictions that distinguish market power from production characteristics.

Our analysis of digitalization and markups is closely related to Calligaris et al. (2018), who examine firm-level markups across 26 countries over 2001–2014. They find that markups are higher in digitally intensive sectors and that the gap between more and less digitally intensive sectors widened over this period. Their estimates follow the production-based approach of De Loecker and Warzynski (2012), which

relies on correctly identified output elasticities. Subsequent research shows that omitting factor-biased productivity can distort these elasticities and the resulting markup estimates (Doraszelski and Jaumandreu, 2019; Demirer, 2026). This markup estimation strategy can be sensitive to misspecification of factor-biased productivity. We address this issue by embedding both labor-augmenting and Hicks-neutral productivity into the production framework. This allows us to jointly examine how firm-level digitalization drives the evolution of heterogeneous productivity components and firm markups.

Measurement of digitalization creates a second challenge. Many studies use one indicator, such as ICT spending or IT capital, to represent digital transformation (Bresnahan et al., 2002; Brynjolfsson and Hitt, 1996; Bertschek and Kaiser, 2004; Hempell, 2005; Baily and Lawrence, 2001; Stiroh, 1998, 2002; Morrison, 1997; Gordon, 2000; Jorgenson and Stiroh, 1999; Oliner and Sichel, 2000; Jorgenson, 2001). Earlier work notes the limitations of such measures (Black and Lynch, 2001; Greenan and Mairesse, 2000; Loveman, 1994). The ESEE allows a broader approach. Following Añón Higón and Bonvin (2024), we combine 13 variables into an overall index and four component indices. The components capture IT technologies, digital human capital, automation, and interaction with stakeholders.

3 Data

We use the Encuesta Sobre Estrategias Empresariales (ESEE), an annual panel of Spanish manufacturing firms covering 2000–2019. The survey reports output, inputs, digital technology variables, changes in input and output prices, and other firm characteristics. After cleaning and applying the lag requirements, the overall PC estimation sample with the overall index contains 2,213 firms and 17,087 firm-year observations. We retain firms with at least three usable observations in the relevant estimation sample. Sample size varies across specifications because of lag requirements and data availability. We classify firms with fewer than 200 workers as small and firms with at least 200 workers as large. Table 2 reports the corresponding numbers of observations and firms.

The ESEE reports changes in output and materials prices rather than price levels. Following Jaumandreu and Lin (2018), we construct Paasche price indices for each firm. Separate indices are constructed for output and intermediate inputs. We use these indices to convert nominal production and intermediate input expenditures into real quantities.

Real output is nominal production of goods and services divided by the firm-specific output price index. We construct real capital using the perpetual inventory method with annual depreciation $d_K = 0.15$. With PI_t equal to the investment-price index divided by 100, the implemented recursion is

$$K_{it} = (1 - d_K)K_{i,t-1} + \frac{I_{i,t-1}}{PI_t}.$$

Investment made in period $t - 1$ enters capital used in period t ; the code deflates that lagged investment by the current-year investment-price index. At the start of each available consecutive-year spell, we initialize capital using positive reported real net capital when available.

Materials are measured by total intermediate consumption. This includes raw materials, energy, and purchased services. We deflate nominal expenditures by the firm-specific intermediate input price index. The resulting measure captures changes in real materials rather than changes in expenditure caused only by prices.

Labor is measured by total hours worked. The hourly wage equals total labor compensation divided by total hours. This separates the quantity of labor from the price of labor services. We do not use the wage as a measure of labor efficiency. Labor efficiency is recovered from the model under the maintained identification assumptions.

3.1 Digitalization Indices

We construct a multidimensional digitalization index following Añón Higón and Bonvin (2024). The index contains 13 variables grouped into four components. The IT technology component includes investment in computer equipment and the use of external software programming services. The digital human capital component measures external expenditure on employee training in computing and information technologies. The automation component covers the use of industrial robots, computer-aided design, flexible manufacturing systems, local area networks, and CNC machine tools. The stakeholder interaction component includes ownership of an internet domain, a company webpage, online sales to businesses and consumers, and purchases from suppliers through the internet. We include CNC machine tools to reflect the ESEE manufacturing setting.

We construct each digitalization measure according to the type of survey variable. Monetary variables are converted to real values, while categorical variables are converted into indicator variables. The two monetary variables are investment in computer equipment and external expenditure on IT training. The ESEE reports computer equipment investment as a percentage of total capital goods investment. We convert this percentage into an absolute flow by multiplying it by total capital goods investment. We then convert both monetary flows into stocks using the perpetual inventory method with a 15% depreciation rate. Each stock is divided by its industry and year mean. We assign the resulting relative stock to a decile and rescale it as $(d - 1)/9$, where $d \in \{1, \dots, 10\}$. The resulting scores lie between zero and one.

The categorical variables are survey adoption codes rather than ordered measures of intensity. Six variables are observed every four years: robots, CAD, flexible manufacturing systems, LAN, CNC machine tools, and external software programming services. After converting these variables to binary indicators, we carry an observed response forward for at most three years within a firm. This fills the years between survey waves. It does not replace an explicit missing response or nonresponse code. The five stakeholder variables are observed annually and remain binary after recoding.

The overall index is the unweighted sum of the 13 transformed variables. At the final aggregation stage, we set a missing transformed component to zero. We then normalize the raw sum to a scale from zero to one using its minimum and maximum values. A higher value indicates a broader and more intensive digital presence. We construct the four component indices in the same way.

Finally, we classify firms as either highly or low digitalized based on the taxonomy proposed by Calvino et al. (2018).¹ We further stratify our sample by size, defining small firms as those with fewer than 200 employees, and large firms as those with 200 employees or more. We conduct our estimations on these distinct sub-samples as well as on the pooled full sample.

3.2 Summary Statistics

Table 1 presents summary statistics for output, inputs, and prices across 17,087 firm-year observations. Mean real gross output is 51.3 million euros, compared with a median of 10.1 million euros and a standard deviation of 102.9 million euros. Real capital and materials average 12.4 million and 36.2 million euros, respectively, with corresponding medians of 2.0 million and 6.5 million euros. Annual labor input averages approximately 314 thousand hours per firm, with a median of 101 thousand hours and a standard deviation of 483.5 thousand hours. Across these variables, means substantially exceed medians, and standard deviations exceed means, indicating considerable heterogeneity in production scale and suggesting right-skewed distributions.

Price indices show less relative dispersion than output and input quantities. The output price index averages 1.008, with a standard deviation of 0.094, whereas the materials price index averages 1.038, with a standard deviation of 0.153. Nominal hourly wages average 19.29 euros, with a median of 18.35 euros; the middle half of observations lies between 13.68 and 23.72 euros per hour. The capital investment-price index averages 0.831, with a standard deviation of 0.089. These statistics describe variation across the pooled firm-year sample and therefore reflect both differences across firms and changes over time.

Table 1: Summary statistics for output, inputs, and prices

Variable	Observations	Mean	SD	P25	Median	P75
Real gross output (million)	17,087	51.326	102.912	2.396	10.083	47.380
Real capital (million)	17,087	12.412	25.570	0.433	2.042	11.004
Labor (thousand hours)	17,087	313.972	483.489	41.000	101.000	378.833
Real materials (million)	17,087	36.206	74.528	1.400	6.495	32.338
Output price index	17,087	1.008	0.094	0.953	0.996	1.051
Materials price index	17,087	1.038	0.153	0.948	1.014	1.112
Nominal hourly wage (hour)	17,087	19.287	7.458	13.678	18.347	23.724
Capital investment-price index	17,087	0.831	0.089	0.749	0.862	0.912

Notes: Quantities, input/output prices and wages use the coherent-winsorization overall PC panel. Real output and materials equal reconstructed nominal values divided by their prices; wages equal personnel cost divided by hours. The final row retains the source capital-investment deflator on those same firm-years. Monetary values retain source units, with display scaling indicated in row labels.

Table 2 reports summary statistics for the digitalization measures together with sample coverage. The overall estimation with the overall index uses 17,087 firm-year observations from 2,213 firms. The estimation with the four component indices uses 15,874 observations from 2,140 firms. The table reports the corresponding counts for each group. Differences across specifications reflect sample cleaning, data

¹Following Calvino et al. (2018), low digitalized firms are those operating in the metals and metal products, non-metallic minerals, chemical products, food, beverages and tobacco, and textile, leather, and footwear industries. All remaining firms are classified as highly digitalized.

availability, and lag requirements.

Small firms employ fewer than 200 workers, while large firms employ at least 200. The low digital industry group includes metal and metal products, nonmetallic minerals, chemical products, food, drink and tobacco, and textiles, leather and shoes. The high digital group includes agricultural and industrial machinery, electrical goods, transport equipment, timber and furniture, and paper and printing. We exclude other manufacturing from these industry comparisons. These groups classify industries rather than individual firms based on their own digitalization scores.

The mean overall digitalization index is 0.356 in the overall sample. The mean is 0.291 for small firms and 0.526 for large firms. Large firms also have higher means for all four component indices. The largest difference is in digital human capital, with means of 0.151 for small firms and 0.611 for large firms. Differences across industry groups are smaller. The mean overall index is 0.334 in low-digital industries and 0.394 in high-digital industries. These comparisons are descriptive and are based on the relevant estimation samples.

The component scores combine binary adoption measures with transformed expenditure stocks. A zero score for digital human capital does not necessarily imply that a firm provides no training. It may reflect the lowest relative stock category or the treatment of a missing component. The scores therefore measure relative digitalization rather than physical technology stocks or adoption probabilities.

Table 2: Digitalization summary statistics and estimation-sample coverage

Measure	Observations	Firms	Mean	SD	P25	Median	P75
<i>Overall</i>							
Overall index	17,087	2,213	0.356	0.208	0.187	0.346	0.508
IT technologies	15,874	2,140	0.431	0.325	0.143	0.375	0.714
Digital human capital	15,874	2,140	0.276	0.369	0.000	0.000	0.500
Automation	15,874	2,140	0.388	0.324	0.200	0.400	0.600
Stakeholder-facing	15,874	2,140	0.328	0.232	0.200	0.400	0.400
<i>Small firms</i>							
Overall index	12,246	1,689	0.291	0.181	0.154	0.275	0.421
IT technologies	11,389	1,625	0.334	0.284	0.100	0.250	0.556
Digital human capital	11,389	1,625	0.151	0.273	0.000	0.000	0.250
Automation	11,389	1,625	0.314	0.293	0.000	0.200	0.600
Stakeholder-facing	11,389	1,625	0.297	0.224	0.200	0.200	0.400
<i>Large firms</i>							
Overall index	4,343	546	0.526	0.183	0.385	0.529	0.663
IT technologies	3,988	525	0.691	0.282	0.444	0.786	0.944
Digital human capital	3,988	525	0.611	0.385	0.286	0.714	1.000
Automation	3,988	525	0.580	0.323	0.400	0.600	0.800
Stakeholder-facing	3,988	525	0.414	0.247	0.200	0.400	0.600
<i>Low-digital industries</i>							
Overall index	10,610	1,374	0.334	0.204	0.171	0.321	0.485
IT technologies	9,871	1,328	0.425	0.325	0.143	0.375	0.714
Digital human capital	9,871	1,328	0.270	0.368	0.000	0.000	0.500
Automation	9,871	1,328	0.348	0.315	0.000	0.200	0.600
Stakeholder-facing	9,871	1,328	0.315	0.229	0.200	0.400	0.400
<i>High-digital industries</i>							
Overall index	5,883	789	0.394	0.208	0.240	0.387	0.538
IT technologies	5,459	762	0.442	0.325	0.167	0.389	0.750
Digital human capital	5,459	762	0.288	0.367	0.000	0.000	0.600
Automation	5,459	762	0.459	0.327	0.200	0.400	0.800
Stakeholder-facing	5,459	762	0.351	0.236	0.200	0.400	0.400

Notes: Statistics use the corrected PC estimation panels. Digital indices retain within-sample/year 2.5th–97.5th percentile winsorization. Nominal accounts, prices and quantities are linked through the coherent preprocessing described in the text. Subgroup firm counts may overlap.

4 Model

This section first specifies a production technology with labor-augmenting and Hicks-neutral productivity. We then present the PC benchmark followed by its extension to IPC. Firms are price takers in the labor and materials markets in both cases. Only the output-market assumption changes. The production technology is shared across the two cases. The IPC model additionally requires the dynamic moment and covariance restrictions stated below.

4.1 Technology

Firm i produces gross output at time t using predetermined capital K_{it} , freely variable labor L_{it} , and freely variable materials M_{it} . The production technology is given by

$$Y_{it} = Y_{it}^* e^{\eta_{it}} = F(K_{it}, e^{\omega_{L,it}} L_{it}, M_{it}) e^{\omega_{N,it} + \eta_{it}}, \quad (1)$$

where η_{it} is a productivity shock unobserved by all agents, $\omega_{L,it}$ is labor-augmenting productivity, and $\omega_{N,it}$ is Hicks-neutral productivity. Both $\omega_{N,it}$ and $\omega_{L,it}$ are known to the firm when inputs are chosen, whereas η_{it} is realized after inputs are chosen. We assume $\mathbb{E}[\eta_{it} \mid \mathcal{I}_{it}] = 0$ and define

$$\theta \equiv \mathbb{E}[e^{\eta_{it}} \mid \mathcal{I}_{it}], \quad (2)$$

which is assumed to be constant across firms and over time.

We use the lower-case letters as natural logarithms of their upper-case counterparts, and write $l_{it}^* \equiv l_{it} + \omega_{L,it}$ for effective labor in logs. The firm's information set $\mathcal{I}_{it}$ contains K_{it} , $\omega_{L,it}$, $\omega_{N,it}$, all input prices, and the entire history of these variables.

We assume $\ln F(\cdot)$ to be translog:

$$\begin{aligned} \ln F(K_{it}, L_{it}^*, M_{it}) = & \alpha_0 + \beta_K k_{it} + \beta_L l_{it}^* + \beta_M m_{it} + \frac{1}{2} \beta_{KK} k_{it}^2 + \frac{1}{2} \beta_{LL} (l_{it}^*)^2 + \frac{1}{2} \beta_{MM} m_{it}^2 \\ & + \beta_{KL} k_{it} l_{it}^* + \beta_{KM} k_{it} m_{it} + \beta_{ML} m_{it} l_{it}^*. \end{aligned} \quad (3)$$

Following Doraszelski and Jaumandreu (2019) and Malikov et al. (2024) (MZZ), we impose separability of capital from the two variable inputs, which requires $\beta_{KL} = \beta_{KM} = 0$. We further impose homogeneity of degree ν in effective labor and materials, which requires $\beta_{LL} + \beta_{ML} = 0$ and $\beta_{MM} + \beta_{ML} = 0$. Denoting $\delta \equiv \beta_{ML}$, these restrictions imply $\beta_{LL} = \beta_{MM} = -\delta$, and we define the degree of homogeneity as

$$\nu \equiv \beta_L + \beta_M. \quad (4)$$

Under these restrictions, the technology can be written as

$$y_{it}^* = \alpha_0 + \beta_K k_{it} + \frac{1}{2} \beta_{KK} k_{it}^2 + \beta_M m_{it} + \beta_L (l_{it} + \omega_{L,it}) - \frac{1}{2} \delta (m_{it} - l_{it} - \omega_{L,it})^2 + \omega_{N,it}. \quad (5)$$

4.2 Output elasticities

Differentiating (5) at fixed $\omega_{L,it}$ gives the observation-specific output elasticities

$$\varepsilon_{it}^L = \beta_L + \delta (m_{it} - l_{it} - \omega_{L,it}), \quad (6)$$

$$\varepsilon_{it}^M = \beta_M - \delta (m_{it} - l_{it} - \omega_{L,it}), \quad (7)$$

$$\varepsilon_{it}^K = \beta_K + \beta_{KK} k_{it}. \quad (8)$$

Homogeneity implies that the elasticity with respect to a proportional change in labor and materials is a constant,

$$\varepsilon_{it}^L + \varepsilon_{it}^M = \nu. \quad (9)$$

4.3 PC model

We use the PC model in which the firm takes the output price P_{it}^Y as exogenously given as the benchmark. Our PC estimation follows the approach of Malikov et al. (2024) (MZZ). Since η affects realized output, the first-order conditions (FOCs) of expected profit maximization over L_{it} and M_{it} imply

$$P_{it}^Y \theta \varepsilon_{it}^L \frac{e^{y_{it}^*}}{L_{it}} = P_{it}^L, \quad (10)$$

$$P_{it}^Y \theta \varepsilon_{it}^M \frac{e^{y_{it}^*}}{M_{it}} = P_{it}^M. \quad (11)$$

We now define the revenue shares and their sum,

$$s_{it}^L = \frac{P_{it}^L L_{it}}{P_{it}^Y Y_{it}}, \quad s_{it}^M = \frac{P_{it}^M M_{it}}{P_{it}^Y Y_{it}}, \quad R_{it} = s_{it}^L + s_{it}^M. \quad (12)$$

Dividing (10) and (11) by revenue $P_{it}^Y Y_{it}$, adding, and using (9) gives the share equation

$$\ln R_{it} = \ln(\theta \nu) - \eta_{it}. \quad (13)$$

Taking the ratio of (10) and (11) gives

$$\frac{P_{it}^L L_{it}}{P_{it}^M M_{it}} = \frac{\varepsilon_{it}^L}{\varepsilon_{it}^M} = \frac{\beta_L + \delta(m_{it} - l_{it} - \omega_{L,it})}{\beta_M - \delta(m_{it} - l_{it} - \omega_{L,it})}. \quad (14)$$

The output price, Hicks-neutral productivity, θ , and η_{it} all cancel.

Solving (14) gives

$$\delta(m_{it} - l_{it} - \omega_{L,it}) = \beta_M SC_{it}^L - \beta_L SC_{it}^M = \nu SC_{it}^L - \beta_L, \quad (15)$$

and hence the labor-augmenting-productivity term can be expressed as

$$\omega_{L,it} = m_{it} - l_{it} + \frac{\beta_L}{\delta} - \frac{\nu}{\delta} SC_{it}^L. \quad (16)$$

where labor share of total variable cost is

$$SC_{it}^L = \frac{P_{it}^L L_{it}}{P_{it}^L L_{it} + P_{it}^M M_{it}}, \Rightarrow SC_{it}^M = 1 - SC_{it}^L. \quad (17)$$

Substituting (15) into (6)–(7) yields the identities

$$\varepsilon_{it}^L = \nu SC_{it}^L, \quad \varepsilon_{it}^M = \nu SC_{it}^M. \quad (18)$$

Equations (16) and (18) are implications of the restricted translog with two variable inputs. They hold in both market environments studied below, because a markup common to the two conditions cancels from the ratio. This common inversion is the point at which our extension connects to the competitive benchmark.

Substituting (15) into (5) and collecting terms concentrates effective labor out of the production function:

$$y_{it}^* = \kappa_0 + \beta_K k_{it} + \frac{1}{2} \beta_{KK} k_{it}^2 + \nu m_{it} - \frac{\nu^2}{2\delta} (SC_{it}^L)^2 + \omega_{N,it}, \quad \kappa_0 = \alpha_0 + \frac{\beta_L^2}{2\delta}. \quad (19)$$

Equation (19) expresses output in terms of ν , δ , and the composite intercept κ_0 .

To estimate the competitive benchmark model outlined above, we adopt the following three-step procedure.²

4.3.1 Step 1: Estimation of θ , ν and prediction of η_{it}

Let N_{obs} denote the number of firm-years of observations. Because $\mathbb{E}[\eta_{it}] = 0$, the sample analogues of (13) are

$$\hat{\eta}_{it} = \overline{\ln R} - \ln R_{it}, \quad (20)$$

$$\hat{\theta} = \frac{1}{N_{\text{obs}}} \sum_{it} \exp(\hat{\eta}_{it}), \quad (21)$$

$$\hat{\nu} = \frac{\exp(\overline{\ln R})}{\hat{\theta}}. \quad (22)$$

The first step therefore identifies $\nu = \beta_L + \beta_M$ and θ and gives a prediction of η_{it} .

4.3.2 Step 2: Prediction of labor-augmenting productivity

Let $\mathbf{D}_{it}$ collect the firm's digitalization measures. We assume that labor-augmenting productivity follows the following law of motion

$$\omega_{L,it} = \rho_L \omega_{L,i,t-1} + \gamma_L' \mathbf{D}_{i,t-1} + \xi_{L,it}, \quad \mathbb{E}[\xi_{L,it} \mid \mathcal{I}_{i,t-1}] = 0. \quad (23)$$

The equation contains no intercept. This is standard in the IO productivity literature (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Akerberg et al., 2015; Zhao et al., 2024)

Step 2 estimates the law of motion in (23). Since $\omega_{L,it}$ is unobserved, we replace it using the labor

²For details on the estimation of the restricted translog model with factor-biased productivity, see Malikov et al. (2024).

productivity proxy in (16). Substituting the proxy at t and $t - 1$ gives

$$\begin{aligned}\xi_{L,it} = & (m_{it} - l_{it}) + \frac{\beta_L}{\delta} - \frac{\nu}{\delta} SC_{it}^L \\ & - \rho_L \left[(m_{i,t-1} - l_{i,t-1}) + \frac{\beta_L}{\delta} - \frac{\nu}{\delta} SC_{i,t-1}^L \right] - \gamma'_L \mathbf{D}_{i,t-1}.\end{aligned}\quad (24)$$

This equation serves as a proxy for the innovation term. We estimate the parameter vector $\left(\frac{\beta_L}{\delta}, \frac{\nu}{\delta}, \rho_L, \gamma'_L\right)'$ by nonlinear GMM. The instrument vector is

$$\mathbf{Q}_{L,it} = (1, m_{i,t-1} - l_{i,t-1}, SC_{i,t-1}^L, \mathbf{D}'_{i,t-1})', \quad (25)$$

which contains a constant, the lagged input ratio, the lagged labor cost share, and the lagged digitalization variables. The corresponding moment conditions are

$$\mathbb{E} \left[\mathbf{Q}_{L,it} \xi_{L,it} \left(\frac{\beta_L}{\delta}, \frac{\nu}{\delta}, \rho_L, \gamma_L \right) \right] = \mathbf{0}. \quad (26)$$

The number of instruments equals the number of parameters in this block. Step 2 is therefore exactly identified under the maintained exclusion and rank conditions.

Since $\xi_{L,it}$ does not contain Y_{it} , the variables dated $t - 1$ in (25) are valid instruments under the maintained information assumptions. Thus Step 2 provides estimates of β_L/δ and ν/δ , together with ρ_L and γ_L . Step 1 has already identified $\nu = \beta_L + \beta_M$. We can therefore recover the production function effects as

$$\widehat{\delta} = \frac{\widehat{\nu}}{(\widehat{\nu}/\widehat{\delta})}, \quad \widehat{\beta}_L = (\widehat{\beta}_L/\widehat{\delta})\widehat{\delta}, \quad \widehat{\beta}_M = \widehat{\nu} - \widehat{\beta}_L. \quad (27)$$

Thus, the first two steps identify δ , β_L , and β_M , as well as the persistence and digitalization parameters in the law of motion for labor-augmenting productivity.

4.3.3 Step 3: Hicks-neutral productivity

Hicks-neutral productivity dynamics follows

$$\omega_{N,it} = \rho_{0N} + \rho_N \omega_{N,i,t-1} + \gamma'_N \mathbf{D}_{i,t-1} + \xi_{N,it}, \quad \mathbb{E}[\xi_{N,it} | \mathcal{I}_{i,t-1}] = 0. \quad (28)$$

Two representations of $\omega_{N,it}$ are available. From the production function,

$$y_{it}^\dagger \equiv y_{it} - \beta_M m_{it} - \beta_L (l_{it} + \omega_{L,it}) + \frac{1}{2} \delta (m_{it} - l_{it} - \omega_{L,it})^2 = \alpha_0 + \beta_K k_{it} + \frac{1}{2} \beta_{KK} k_{it}^2 + \omega_{N,it} + \eta_{it}. \quad (29)$$

Using (29) we express

$$\omega_{N,it} = y_{it}^\dagger - [\alpha_0 + \beta_K k_{it} + \frac{1}{2} \beta_{KK} k_{it}^2 + \eta_{it}] \quad (30)$$

Alternatively, we can use materials demand to proxy for unobserved Hicks-neutral productivity. Solving

(11) for $\omega_{N,it}$ gives the materials-demand inversion

$$\begin{aligned} m_{it}^\dagger &\equiv (p_{it}^M - p_{it}^Y) - \ln \theta - \ln[\beta_M - \delta(m_{it} - l_{it} - \omega_{L,it})] + (1 - \beta_M) m_{it} \\ &\quad - \beta_L(l_{it} + \omega_{L,it}) + \frac{1}{2}\delta(m_{it} - l_{it} - \omega_{L,it})^2 = \alpha_0 + \beta_K k_{it} + \frac{1}{2}\beta_{KK} k_{it}^2 + \omega_{N,it}, \end{aligned} \quad (31)$$

where p_{it}^M and p_{it}^Y are log input and output prices. We can use (31) to express $\omega_{N,it}$ which is

$$\omega_{N,it} = m_{it}^\dagger - [\alpha_0 + \beta_K k_{it} + \frac{1}{2}\beta_{KK} k_{it}^2]. \quad (32)$$

The inversion (32) does not contain Y_{it} and is therefore free of η_{it} , which is precisely why it, rather than (30), is used to replace lagged Hicks-neutral productivity. Substituting $\omega_{N,i,t-1} = m_{i,t-1}^\dagger - \alpha_0 - \beta_K k_{i,t-1} - \frac{1}{2}\beta_{KK} k_{i,t-1}^2$ into (28) and then into (29) gives

$$\begin{aligned} y_{it}^\dagger &= \pi_0 + \beta_K k_{it} + \frac{1}{2}\beta_{KK} k_{it}^2 + \rho_N \left[m_{i,t-1}^\dagger - \beta_K k_{i,t-1} - \frac{1}{2}\beta_{KK} k_{i,t-1}^2 \right] \\ &\quad + \gamma'_N \mathbf{D}_{i,t-1} + \xi_{N,it} + \eta_{it}, \end{aligned} \quad (33)$$

where $\pi_0 = (1 - \rho_N)\alpha_0$. This specification appears in the Doraszelski and Jaumandreu (2019), Malikov et al. (2024). We estimate this equation by profiled nonlinear least squares in $(\beta_K, \beta_{KK}, \pi_0, \rho_N, \gamma_N)$, holding the step-1 and step-2 estimates fixed. Because the composite error is $\xi_{N,it} + \eta_{it}$ and k_{it} is predetermined, no additional instruments are required. In particular, competitive Step 3 has no $\mathbf{Q}_{N,it}$ instrument vector: it is profiled nonlinear least squares, not an IV or GMM. Standard errors for the full sequence are obtained by a firm-level block bootstrap in which all three steps are re-estimated.

The two productivity series are recovered observation by observation as

$$\widehat{\omega}_{L,it} = m_{it} - l_{it} + \frac{\widehat{\beta}_L}{\widehat{\delta}} - \frac{\widehat{\nu}}{\widehat{\delta}} SC_{it}^L, \quad (34)$$

$$\widehat{\omega}_{N,it} = \widehat{m}_{it}^\dagger - [\widehat{\alpha}_0 + \widehat{\beta}_K k_{it} + \frac{1}{2}\widehat{\beta}_{KK} k_{it}^2] \quad (35)$$

4.4 IPC model

The IPC extension retains exactly the same output-shock structure as the PC benchmark. The firm chooses labor and materials before η_{it} is realized, using the expected profit maximization behavior under the conditional expectation $\theta = \mathbb{E}[e^{\eta_{it}} | \mathcal{I}_{it}]$.

This is where our identification strategy departs from the existing IPC formulation. The revenue-share system alone cannot separately identify the technology parameter ν and the firm-level markup μ_{it} . Rather than identifying them jointly from this system, we first identify ν from the quantity side of the model and then use the revenue-share equation to construct the firm-year markup signal. Section 5.5 provides the full identification argument.

4.4.1 FOCs and markup identification

To model markup, we assume that firm i faces the inverse demand function $P_{it} = P(Y_{it}^*, z_{it})$, where z_{it} denotes demand shifters. Because the multiplicative shock is realized after variable inputs are chosen, expected-profit maximization introduces the same factor θ as in the PC benchmark. Writing MC_{it} for marginal cost and markup $\mu_{it} = P_{it}/MC_{it}$, the variable-input first-order conditions of expected profit maximization can be written as

$$\theta MC_{it} \frac{\partial Y_{it}^*}{\partial j_{it}} = P_{it}^j, \quad j \in \{L, M\}. \quad (36)$$

Multiplying equation (36) by $j_{it}/(P_{it}Y_{it}^*)$ gives

$$\theta \frac{MC_{it}}{P_{it}} \left(\frac{\partial Y_{it}^*}{\partial j_{it}} \frac{j_{it}}{Y_{it}^*} \right) = \frac{P_{it}^j j_{it}}{P_{it} Y_{it}^*}. \quad (37)$$

Thus, the observed revenue shares are

$$s_{it}^j \equiv \frac{P_{it}^j j_{it}}{P_{it} Y_{it}} = \theta \frac{\varepsilon_{it}^j}{\mu_{it}} e^{-\eta_{it}}, \quad j \in \{L, M\}. \quad (38)$$

consequently, either variable input provides a noisy signal of the same markup ($\mu_{it} e^{\eta_{it}}$):

$$\frac{\varepsilon_{it}^L}{s_{it}^L} = \frac{\varepsilon_{it}^M}{s_{it}^M} = \frac{\mu_{it}}{\theta} e^{\eta_{it}}. \quad (39)$$

The ratio of the labor and materials conditions removes μ_{it} and η_{it} :

$$\frac{s_{it}^L}{s_{it}^M} = \frac{\varepsilon_{it}^L}{\varepsilon_{it}^M} = \frac{\beta_L + \delta(m_{it} - l_{it} - \omega_{L,it})}{\beta_M - \delta(m_{it} - l_{it} - \omega_{L,it})}. \quad (40)$$

This is the same as (14). Hence the labor-augmenting productivity inversion is unchanged. Equation (40) gives

$$\omega_{L,it} = m_{it} - l_{it} + \frac{\beta_L}{\delta} - \frac{\nu}{\delta} SC_{it}^L. \quad (41)$$

This is identical to (16). As before, using $\nu = \beta_L + \beta_M$ gives

$$\varepsilon_{it}^L = \nu SC_{it}^L, \quad \varepsilon_{it}^M = \nu SC_{it}^M. \quad (42)$$

Adding the two observed shares gives

$$R_{it} \equiv s_{it}^L + s_{it}^M = \theta \frac{\nu}{\mu_{it}} e^{-\eta_{it}}. \quad (43)$$

Therefore,

$$RR_{it} \equiv \ln(\theta\nu) - \ln R_{it} = \ln \mu_{it} + \eta_{it}. \quad (44)$$

which is different from (12) because of the presence of markup μ_{it} .

5 Identification and Estimation

5.1 Identification of Hicks-neutral productivity

Combining equation (19) with $y_{it} = y_{it}^* + \eta_{it}$ gives

$$y_{it} = \kappa_0 + \beta_K k_{it} + \frac{1}{2}\beta_{KK} k_{it}^2 + \nu m_{it} - \left(\frac{\nu^2}{2\delta}\right) (SC_{it}^L)^2 + \omega_{N,it} + \eta_{it}. \quad (45)$$

Hicks-neutral productivity follows

$$\omega_{N,it} = \pi_N + \rho_N \omega_{N,i,t-1} + \gamma'_N \mathbf{D}_{i,t-1} + \xi_{N,it}, \quad \mathbb{E}[\xi_{N,it} \mid \mathcal{I}_{i,t-1}] = 0. \quad (46)$$

Quasi differencing equation (45) gives

$$\begin{aligned} u_{N,it} = & y_{it} - \pi_0 - \rho_N y_{i,t-1} - \beta_K (k_{it} - \rho_N k_{i,t-1}) \\ & - \frac{1}{2}\beta_{KK} (k_{it}^2 - \rho_N k_{i,t-1}^2) - \nu (m_{it} - \rho_N m_{i,t-1}) \\ & + \left(\frac{\nu^2}{2\delta}\right) [(SC_{it}^L)^2 - \rho_N (SC_{i,t-1}^L)^2] - \gamma'_N \mathbf{D}_{i,t-1}, \end{aligned} \quad (47)$$

where $\pi_0 = \pi_N + (1 - \rho_N)\kappa_0$. The composite disturbance is

$$u_{N,it} = \xi_{N,it} + \eta_{it} - \rho_N \eta_{i,t-1}. \quad (48)$$

The Stage 1 parameter vector is

$$\boldsymbol{\theta}_N = (\pi_0, \beta_K, \beta_{KK}, \nu, \delta, \rho_N, \gamma_N)'. \quad (49)$$

We estimate this block by nonlinear GMM using

$$\mathbb{E}[\mathbf{Z}_{N,it} u_{N,it}] = \mathbf{0}, \quad (50)$$

where $\mathbf{Z}_{N,it}$ contains predetermined and sufficiently dated variables that are orthogonal to $\xi_{N,it}$, η_{it} , and $\eta_{i,t-1}$. In particular, the presence of $\eta_{i,t-1}$ in equation (48) requires the timing of the instruments to be chosen accordingly.

The common instrument block uses capital dated $t - 1$ and its square, together with the $t - 2$ labor cost share, digitalization variables, and materials. Under the stated timing assumptions, lagged capital and its square are orthogonal to both η_{it} and $\eta_{i,t-1}$ and to the current Hicks-neutral innovation $\xi_{N,it}$.

$$\mathbf{Z}_{N,it}^0 = \left(1, (SC_{i,t-2}^L)^2, \mathbf{D}_{i,t-2}', m_{i,t-2}, k_{i,t-1}, k_{i,t-1}^2\right)', \quad (51)$$

and $\mathbf{Z}_{N,it}$ stacks $\mathbf{Z}_{N,it}^0$ with the sample-specific instruments in Table 3.

Table 3: Hicks-neutral productivity with sample-specific instruments

Sample	Sample-specific instruments added to $\mathbf{Z}_{N,it}^0$	Moments / free parameters	
		Composite	Four components
Overall	$R_{i,t-2}, ICT_{i,t-2}$	8 / 7	11 / 10
Small firms	$SC_{i,t-2}^L, R_{i,t-2}$	8 / 7	11 / 10
Large firms	$s_{i,t-2}^L, \ln(\text{effective hours}_{i,t-2})$	8 / 7	11 / 10
Low-digital industries	$R_{i,t-2}, ICT_{i,t-2}, \mathbf{1}\{\text{metal products}\}_i$	9 / 7	12 / 10
High-digital industries	$s_{i,t-2}^L, k_{i,t-1}m_{i,t-2}$	8 / 7	11 / 10

Notes: Every sample uses the common instruments $\mathbf{Z}_{N,it}^0$ in (51). $R_{i,t-2} = s_{i,t-2}^L + s_{i,t-2}^M$ is the lagged sum of the labor and materials expenditure shares of revenue; $s_{i,t-2}^L$ is the lagged labor expenditure share of revenue; $ICT_{i,t-2}$ is the lagged ICT index built from the IT-technology, digital-human-capital and stakeholder-facing items of the digitalization index; $\ln(\text{effective hours}_{i,t-2})$ is lagged log total effective hours; $\mathbf{1}\{\text{metal products}\}_i$ indicates the metal and metal products industry within the low-digital sample. In the component specification $\mathbf{D}_{i,t-2}$ contains the four lagged component indices and the sample-specific instruments are unchanged.

After estimating the technology parameters, we construct the firm-year Hicks-neutral productivity signal

$$H_{it} \equiv y_{it} - \left[\hat{\kappa}_0 + \hat{\beta}_K k_{it} + \frac{1}{2} \hat{\beta}_{KK} k_{it}^2 + \hat{\nu} m_{it} - \left(\frac{\hat{\nu}^2}{2\hat{\delta}} \right) (SC_{it}^L)^2 \right]. \quad (52)$$

By construction,

$$H_{it} = \omega_{N,it} + \eta_{it}. \quad (53)$$

Thus, Stage 2 identifies the technology and produces a signal of Hicks-neutral productivity (confounded with η_{it}). It does not separate $\omega_{N,it}$ from η_{it} .

5.2 Identification of labor-augmenting productivity

Stage 2 uses the Stage 1 Hicks-neutral estimates $(\hat{\nu}, \hat{\delta})$ as fixed inputs when estimating labor-augmenting productivity dynamics. Labor-augmenting productivity follows the controlled first-order Markov process

$$\omega_{L,it} = \rho_L \omega_{L,i,t-1} + \gamma_L' \mathbf{D}_{i,t-1} + \xi_{L,it}, \quad \mathbb{E}[\xi_{L,it} | \mathcal{I}_{i,t-1}] = 0. \quad (54)$$

Evaluated at the Stage 1 estimates of ν and δ , it holds in period t ,

$$\omega_{L,it} = (m_{it} - l_{it}) + \frac{\beta_L}{\hat{\delta}} - \frac{\hat{\nu}}{\hat{\delta}} SC_{it}^L, \quad (55)$$

$$\begin{aligned} (m_{it} - l_{it}) - \frac{\hat{\nu}}{\hat{\delta}} SC_{it}^L = & -(1 - \rho_L) \frac{\beta_L}{\hat{\delta}} + \rho_L \left[(m_{i,t-1} - l_{i,t-1}) - \frac{\hat{\nu}}{\hat{\delta}} SC_{i,t-1}^L \right] \\ & + \gamma_L' \mathbf{D}_{i,t-1} + \xi_{L,it}. \end{aligned} \quad (56)$$

Solving (56) for the innovation gives the residual

$$\begin{aligned}\xi_{L,it} = & (m_{it} - l_{it}) - \rho_L(m_{i,t-1} - l_{i,t-1}) + (1 - \rho_L)\frac{\beta_L}{\hat{\delta}} \\ & - \frac{\hat{\nu}}{\hat{\delta}}(SC_{it}^L - \rho_L SC_{i,t-1}^L) - \gamma_L' \mathbf{D}_{i,t-1},\end{aligned}\quad (57)$$

which depends on the parameter vector

$$\boldsymbol{\theta}_L = (\beta_L, \rho_L, \gamma_L')'. \quad (58)$$

We use the instrument vector

$$\mathbf{Z}_{L,it} = \left(1, (m_{i,t-1} - l_{i,t-1}) - \frac{\hat{\nu}}{\hat{\delta}} SC_{i,t-1}^L, \mathbf{D}_{i,t-1}' \right)', \quad (59)$$

with moment conditions

$$\mathbb{E}[\mathbf{Z}_{L,it} \xi_{L,it}] = \mathbf{0}. \quad (60)$$

5.3 Markup dynamics and the second signal

We allow the log markup to follow

$$\ln \mu_{it} = \pi_\mu + \rho_\mu \ln \mu_{i,t-1} + \gamma_\mu D_{i,t-1} + \xi_{\mu,it}, \quad \mathbb{E}[\xi_{\mu,it} \mid \mathcal{I}_{i,t-1}] = 0. \quad (61)$$

Using the Stage 1 estimate of ν , equation (44) implies the markup signal $RR_{it} = \ln(\theta\nu) - \ln R_{it} = \ln \mu_{it} + \eta_{it}$. Because θ is constant, the empirically unadjusted signal is

$$\widetilde{RR}_{it} \equiv \ln \nu - \ln R_{it} = \ln \mu_{it} + \eta_{it} - \ln \theta \quad (62)$$

The Hicks-neutral and markup blocks therefore provide two observed signals with the same transitory output shock:

$$H_{it} = \omega_{N,it} + \eta_{it}, \quad \widetilde{RR}_{it} = (\ln \mu_{it} - \ln \theta) + \eta_{it}. \quad (63)$$

Their difference removes the common error,

$$H_{it} - \widetilde{RR}_{it} = \omega_{N,it} - \ln \mu_{it} + \ln \theta. \quad (64)$$

This relation provides additional information for separating Hicks-neutral productivity from the markup.

5.4 The state-space decomposition

The baseline model retains $H_{it} = \omega_{N,it} + \eta_{it}$ and $\widetilde{RR}_{it} = \widetilde{m}_{it}^\mu + \eta_{it}$ as noisy firm-year signals, where $\widetilde{m}_{it}^\mu \equiv \ln \mu_{it} - \ln \theta$. If one is willing to impose the additional second-moment restrictions in Appendix C, the

two transition equations and two measurement equations can instead be combined in a state-space system to identify η_{it} . Define the state vector

$$\mathbf{x}_{it} = \begin{pmatrix} \omega_{N,it} \\ \widetilde{m}_{it}^\mu \end{pmatrix}. \quad (65)$$

The state equation is

$$\mathbf{x}_{it} = \begin{pmatrix} \pi_N \\ \widetilde{\pi}_\mu \end{pmatrix} + \begin{pmatrix} \rho_N & 0 \\ 0 & \rho_\mu \end{pmatrix} \mathbf{x}_{i,t-1} + \begin{pmatrix} \gamma_N \\ \gamma_\mu \end{pmatrix} D_{i,t-1} + \begin{pmatrix} \xi_{N,it} \\ \xi_{\mu,it} \end{pmatrix}. \quad (66)$$

The measurement system is

$$\begin{pmatrix} H_{it} \\ \widetilde{RR}_{it} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x}_{it} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \eta_{it}. \quad (67)$$

We assume η_{it} has mean zero and is serially independent. It is also orthogonal to the state innovations $\xi_{N,it}$ and $\xi_{\mu,it}$. The state innovations have mean zero and a finite covariance matrix. The law-of-motion parameters and the innovation variances provide the restrictions needed to distinguish the persistent states from the common transitory output shock. Specifically, quasi-differencing each measurement equation by its own transition parameter, as in equation (48), gives a composite residual whose own- and cross-autocovariances identify σ_η^2 , $\text{Var}(\xi_{N,it})$, $\text{Var}(\xi_{\mu,it})$, and $\text{Cov}(\xi_{N,it}, \xi_{\mu,it})$, provided $\rho_N \neq 0$ and $\rho_\mu \neq 0$; this is the rank condition that makes the dynamics of $\omega_{N,it}$, $\ln \mu_{it}$, and η_{it} non-observationally-equivalent. Because the two signals share the same η_{it} , σ_η^2 is in fact overidentified, recoverable from either signal's own lag-one autocovariance or from their lag-one cross-covariance, giving a built-in specification check. Appendix C states this result formally (Proposition B.2) and verifies it numerically.

Under those additional restrictions, the Kalman filter and smoother provide the linear predictions

$$\widehat{\omega}_{N,it} = \mathbb{E}[\omega_{N,it} \mid \mathcal{Y}_i], \quad \widehat{\widetilde{m}}_{it}^\mu = \mathbb{E}[\widetilde{m}_{it}^\mu \mid \mathcal{Y}_i], \quad (68)$$

where $\mathcal{Y}_i$ contains the sequence of the two observed signals for firm i . The common transitory output shock can then be predicted from either measurement equation:

$$\widehat{\eta}_{it} = H_{it} - \widehat{\omega}_{N,it} = \widetilde{RR}_{it} - \widehat{\widetilde{m}}_{it}^\mu. \quad (69)$$

Labor-augmenting productivity does not require the state-space decomposition because both markup and the transitory output shock cancel from the ratio of the two variable input conditions. Its firm-year prediction uses the Stage 1 Hicks-neutral estimates $\widehat{\nu}$ and $\widehat{\delta}$:

$$\widehat{\omega}_{L,it} = m_{it} - l_{it} + \frac{\widehat{\beta}_L}{\widehat{\delta}} - \frac{\widehat{\nu}}{\widehat{\delta}} SC_{it}^L. \quad (70)$$

Thus, the IC model identifies labor-augmenting productivity directly and identifies noisy firm-year signals for

Hicks-neutral productivity and the log markup. Separate firm-year predictions of Hicks-neutral productivity, the markup, and transitory output shock require the additional state-space restrictions in Appendix C.

In summary, the structural GMM steps identify the technology parameters and provide noisy firm-year signals, while the state-space step separates the persistent latent states from the common transitory output shock. Under the second-moment restrictions in Appendix C, Kalman filtering and smoothing recover the smoothed log markup $\hat{m}_{it}^\mu$, the transitory output shock $\hat{\eta}_{it}$, and Hicks-neutral productivity $\hat{\omega}_{N,it}$. Thus, the state-space step converts the noisy structural signals into observation-specific estimates of markup and Hicks-neutral productivity. It does not identify θ or alter the technology estimates, and the decomposition is conditional on the additional second-moment restrictions.

Relation to existing approaches. Our identification strategy is related to several approaches in the production-function literature. Like Doraszelski and Jaumandreu (2019), we allow productivity to have a labor-augmenting component and a Hicks-neutral component. Their identification strategies use external instruments, such as lagged firm-level input prices or demand shifters. Our approach does not require these instruments.

Our identification builds on the system approach of Gandhi et al. (2020) and the non-neutral-productivity framework of Malikov et al. (2024), but addresses the additional identification problem created by imperfect competition. In particular, the revenue-share equation alone cannot separately identify the technology parameter ν and the firm-level markup.

Our approach separates these two objects sequentially and then uses their dynamic structure for signal extraction. After identifying the technology block from the quantity side, the revenue-share and production equations provide noisy signals of markup and Hicks-neutral productivity. A state-space representation, together with Kalman filtering and smoothing, separates the persistent latent states from the common transitory shock and yields observation-specific estimates of markup and Hicks-neutral productivity. This signal-extraction step provides information that the structural GMM equations alone cannot recover.

Other approaches obtain identification under market power by imposing additional structure. Demirer (2026) use a control-function approach to study non-neutral productivity under IPC. Our approach instead retains the proxy-variable structure. Flynn et al. (2019) obtain identification under market power by imposing constant returns to scale. We do not impose this restriction. Returns to scale are estimated along with the other technology parameters. This distinction is relevant for the empirical results reported in the next section.

5.5 Identification Contribution and Relation to Existing Literature

This subsection relates our identification strategy to the existing proxy-variable literature on non-neutral productivity. The key distinction arises under imperfect competition. Under perfect competition, the revenue-share system can identify $\nu = \beta_L + \beta_M$. Under IPC, however, the corresponding equation identifies only ν/μ_{it} , so ν and the firm-level markup cannot be separately identified from that equation alone.

Malikov et al. (2024) address this problem through set identification and also discuss an alternative price-based materials-demand approach. The former provides bounds on the technology parameters, while the latter requires firm-level variation in relative input prices. Our approach instead obtains point identification of ν from the quantity side of the model. Given ν , the revenue-share equation can then be used to recover the markup signal.

We implement the materials-demand approach as a robustness check in Section 7.2. It provides a useful alternative specification, but it relies on firm-level variation in relative input prices.

Our identification strategy avoids both limitations. The key point is that we do not use the revenue-share equation to identify ν . Instead, ν (together with β_K , β_{KK} , and the Hicks-neutral transition parameters) is identified in Section 5.1 from the *quantity side* of the model, i.e., the dynamic production equation (45) combined with the Markovian law of motion for $\omega_{N,it}$, a block that does not involve the markup. Once we obtain ν , we use the revenue-share identity in equation (44) to recover a signal of $\ln \mu_{it}$. The sequential identification also provides the inputs for the state-space analysis in Section 5.4. Once ν is identified, the model yields firm-year signals for Hicks-neutral productivity and the log markup that share the transitory shock η_{it} , as shown in equations (53) and (62). Their own- and cross-autocovariances provide the second-moment information needed for signal extraction. The Kalman filter and smoother then use the joint time-series structure to separate the persistent latent states from the common transitory shock, yielding observation-specific estimates of $\omega_{N,it}$ and $\ln \mu_{it}$. Proposition B.2 provides an internal check on the estimated variance of the common transitory shock, σ_η^2 .

Our approach is related to several strands of the productivity and market-power literature. We follow Doraszelski and Jaumandreu (2018) and Malikov et al. (2024) in separating labor-augmenting from Hicks-neutral productivity, but do not rely on external instruments such as lagged firm-level input prices or demand shifters. Our identification is closer to the system approach of Malikov et al. (2024) and Gandhi et al. (2020), which identifies scalar productivity without price instruments, but we allow for multiple productivity components and imperfect competition and add a state-space layer to recover observation-specific latent states. Demirer (2026) also study non-neutral productivity under market power using a control-function approach, whereas we use a structural proxy-variable approach and obtain point estimates. Finally, Flynn et al. (2019) obtain identification under market power by imposing constant returns to scale. We do not impose this restriction, which is relevant in our application. This is because most sample point estimates in Section 6 indicate decreasing returns to scale, while the large-firm IPC estimate is close to one.

To summarize, our identification strategy has two distinct layers. First, we identify ν from the quantity side of the model and only then use the revenue-share equation to construct the markup signal. This sequencing separates technology identification from markup identification. Second, the state-space model exploits the joint dynamics of the markup and Hicks-neutral-productivity signals. Under the additional second-moment restrictions, Kalman filtering and smoothing separate the common transitory shock and deliver observation-specific estimates of $\ln \mu_{it}$ and $\omega_{N,it}$. The combination of these two steps, rather than a modification of a single revenue-share equation, is the paper’s main identification contribution.

6 Empirical Results

We first analyze the results under perfect competition (PC) then imperfect competition (IPC). Under each section, we analyze the impact of digitalization on productivity, followed by its effects on markups, subgroup differences, and the four distinct components of digitalization.

6.1 Perfect-competition (PC)

In this section we present the results under PC.

6.1.1 Digitalization and productivity

Table 4 reports the effects of digitalization in the PC productivity transition processes, conditional on current productivity. A one-unit increase covers the index’s entire zero-to-one range. A 0.10-point increase corresponds to one tenth of the reported effect. The overall labor-augmenting estimate of 0.037 therefore implies approximately 0.37% higher next-period labor-augmenting productivity. Persistence allows the effect of a sustained increase in adoption to accumulate over time.

Under PC, the overall labor-augmenting estimate is 0.037 with a bootstrap standard error of 0.007. The overall Hicks-neutral estimate is 0.062 with a bootstrap standard error of 0.008. The labor-augmenting estimates are 0.046 for small firms and 0.064 for high-digital industries. These estimates are significant. The estimates for large firms and low-digital industries are not distinguishable from zero. Hicks-neutral estimates are positive and significant in every subgroup.

Table 4: Digitalization effects in productivity transitions under PC

Outcome	Overall	Small	Large	Low	High
Labor-augmenting productivity	0.037*** (0.007)	0.046*** (0.009)	−0.010 (0.015)	0.016 (0.011)	0.064*** (0.012)
Hicks-neutral productivity	0.062*** (0.008)	0.067*** (0.011)	0.033** (0.014)	0.061*** (0.011)	0.062*** (0.014)
Observations	17,087	12,246	4,343	10,610	5,883
Firms	2,213	1,689	546	1,374	789

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 attempts re-estimating all stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference conditions on preprocessing, the fixed instrument sets and the selected solution branch. The index ranges from zero to one. A 0.10-point increase corresponds to one tenth of the reported effect. PC uses one lag and the minimal exactly identified MZZ labor instrument set.

6.1.2 Heterogeneity in the effects of digitalization in the PC

Table 5 tests whether the digitalization effects reported in Table 4 differ across firm-size classifications and between low- and high-digital-intensity industries. Because productivity is estimated allowing for different technological parameters across samples, baseline productivity levels are not directly comparable across subgroups. Consequently, our comparisons focus strictly on the marginal effects of digitalization.

The evidence of heterogeneity is strongest for labor-augmenting productivity. Under PC, the digitalization effect is 0.056 larger for small firms than for large firms, a difference significant at the one percent

level. The small-firm effect is positive and significant, while the large-firm effect is not statistically distinguishable from zero. Across industries, the labor-augmenting effect is 0.048 larger in high-digital-intensity industries than in low-digital-intensity industries.

The PC Hicks-neutral effect is 0.034 larger for small firms than for large firms. The bootstrap standard error of this difference is 0.017. The high-minus-low industry difference is 0.001 with a bootstrap standard error of 0.017.

Table 5: Differences in digitalization effects under PC

Outcome	Small	Large	Small-large	High	Low	High-low
Labor-augmenting productivity	0.046*** (0.009)	-0.010 (0.015)	0.056*** (0.017)	0.064*** (0.012)	0.016 (0.011)	0.048*** (0.016)
Hicks-neutral productivity	0.067*** (0.011)	0.033** (0.014)	0.034** (0.017)	0.062*** (0.014)	0.061*** (0.011)	0.001 (0.017)

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 attempts re-estimating all stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference conditions on preprocessing, the fixed instrument sets and the selected solution branch. Differences use a joint union-firm bootstrap, preserving covariance when firms appear in both groups. Industry digital intensity does not classify individual firms by their own adoption scores. All differences use 499 draws.

6.1.3 Effects of Digitalization Dimensions on Productivity

We next replace the composite index with its four components, described in Section 3, which enter both productivity transitions jointly. Table 6 reports the estimates under PC, with the labor-augmenting process in Panel A and the Hicks-neutral process in Panel B. Because the components enter together, each effect measures the estimated effect of one dimension of digitalization on productivity, holding the other three constant.

PC labor-augmenting effects vary across components. The estimated effect of IT technologies is 0.010 overall. The estimated effect of IT technologies is 0.019 in high-digital industries. Estimates for digital human capital are 0.019 overall and 0.021 for small firms. They are 0.021 in low-digital industries and 0.020 in high-digital industries. Estimates for automation are 0.010 overall and 0.018 in high-digital industries. The estimated effect of automation is 0.019 for small firms. The stakeholder-facing digitalization estimates cannot be distinguished from zero in any sample.

PC Hicks-neutral effects also vary across components. Estimates for IT technologies are 0.011 overall and 0.017 for large firms. The estimated effect of IT technologies is 0.023 in high-digital industries. The estimated effect of digital human capital is 0.019 in high-digital industries. Estimates for automation are 0.018 overall and 0.022 for small firms. The effect is 0.027 in low-digital industries. Estimates for stakeholder-facing digitalization are 0.021 overall and 0.026 for small firms. The estimated effect of stakeholder-facing digitalization is 0.017 in low-digital industries. The estimated effect of stakeholder-facing digitalization is 0.028 in high-digital industries.

The PC component estimates show distinct productivity channels. Digital human capital contributes to labor-augmenting productivity in most samples, while IT technologies and automation are significant in selected samples. Automation and stakeholder-facing digitalization have significant Hicks-neutral effects

in the overall and small-firm samples. In high-digital industries, IT technologies and stakeholder-facing digitalization are significant at five percent or better, whereas digital human capital is significant only at ten percent. These results do not attribute labor-augmenting gains exclusively to digital human capital.

Table 6: Digitalization components and productivity effects under PC

Component	Overall	Small	Large	Low	High
<i>Labor-augmenting productivity</i>					
IT technologies	0.010** (0.005)	0.008 (0.006)	0.002 (0.010)	0.006 (0.006)	0.019*** (0.007)
Digital human capital	0.019*** (0.004)	0.021*** (0.006)	0.008 (0.007)	0.021*** (0.005)	0.020*** (0.007)
Automation	0.010** (0.004)	0.019*** (0.005)	-0.007 (0.007)	0.005 (0.006)	0.018** (0.007)
Stakeholder-facing digitalization	0.007 (0.007)	0.009 (0.008)	0.000 (0.011)	-0.007 (0.009)	0.016 (0.010)
<i>Hicks-neutral productivity</i>					
IT technologies	0.011* (0.005)	0.007 (0.007)	0.017* (0.009)	0.005 (0.007)	0.023*** (0.009)
Digital human capital	0.009 (0.006)	0.008 (0.008)	0.010 (0.009)	0.003 (0.007)	0.019* (0.010)
Automation	0.018*** (0.005)	0.022*** (0.006)	0.002 (0.009)	0.027*** (0.007)	0.000 (0.008)
Stakeholder-facing digitalization	0.021*** (0.007)	0.026*** (0.009)	0.005 (0.012)	0.017* (0.009)	0.028** (0.011)
Observations	15,874	11,389	3,988	9,871	5,459
Firms	2,140	1,625	525	1,328	762

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 attempts re-estimating all stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference conditions on preprocessing, the fixed instrument sets and the selected solution branch. effects describe conditional effects, not independently identified causal effects. Four component indices enter each transition jointly. Bootstrap failures and nonfinite draws are retained in the replication records, rather than replaced by additional attempts. All displayed effects use 499 draws.

6.2 IPC

In this section, we present the results under IPC.

6.2.1 Digitalization and productivity (IC)

Table 7 reports IPC estimates. For the overall sample, the labor-augmenting estimate is 0.055 with a bootstrap standard error of 0.007. For small firms, the labor-augmenting estimate is 0.059 with a bootstrap standard error of 0.008. For large firms, the labor-augmenting estimate is -0.002 with a bootstrap standard error of 0.011. For low-digital industries, the labor-augmenting estimate is 0.050 with a bootstrap standard error of 0.010. For high-digital industries, the labor-augmenting estimate is 0.062 with a bootstrap standard error of 0.017. These estimates are statistically significant and distinguishable from zero. The estimate for large firms is not distinguishable from zero.

For the overall sample, the Hicks-neutral estimate is 0.043 with a bootstrap standard error of 0.008. For small firms, the Hicks-neutral estimate is 0.042 with a bootstrap standard error of 0.010. For large firms, the Hicks-neutral estimate is 0.054 with a bootstrap standard error of 0.026. For low-digital industries, the Hicks-neutral estimate is 0.049 with a bootstrap standard error of 0.010. For high-digital industries, the

Hicks-neutral estimate is 0.019 with a bootstrap standard error of 0.017. The estimate for high-digital industries is not distinguishable from zero.

Table 7: Digitalization effects in productivity and markup transitions under IPC

Outcome	Overall	Small	Large	Low	High
Labor-augmenting productivity	0.055*** (0.007)	0.059*** (0.008)	-0.002 (0.011)	0.050*** (0.010)	0.062*** (0.017)
Hicks-neutral productivity	0.043*** (0.008)	0.042*** (0.010)	0.054** (0.026)	0.049*** (0.010)	0.019 (0.017)
Log-markup transition	0.007** (0.003)	0.013** (0.005)	0.003 (0.007)	0.012*** (0.004)	-0.007 (0.004)
Observations	13,291	9,266	3,415	8,278	4,523
Firms	1,816	1,324	457	1,120	656

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 attempts re-estimating all stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference conditions on preprocessing, the fixed instrument sets and the selected solution branch. effects describe conditional effects, not independently identified causal effects. The index ranges from zero to one. A 0.10-point increase corresponds to one tenth of the reported effect. IPC uses two lags and estimates Hicks-neutral technology first, labor dynamics conditional on that technology second, and markup dynamics third. Finite productivity draws in column order: 499, 499, 499, 499, 499. Finite markup draws: 499, 499, 497, 499, 499. Nonfinite draws remain recorded.

6.2.2 Digitalization and markup dynamics

Table 7 also reports markup-transition effects. For the overall sample, the markup-transition estimate is 0.007 with a bootstrap standard error of 0.003. For small firms, the markup-transition estimate is 0.013 with a bootstrap standard error of 0.005. For large firms, the markup-transition estimate is 0.003 with a bootstrap standard error of 0.007. For low-digital industries, the markup-transition estimate is 0.012 with a bootstrap standard error of 0.004. For high-digital industries, the markup-transition estimate is -0.007 with a bootstrap standard error of 0.004. These estimates are significant and distinguishable from zero. However, the estimates for large firms and high-digital industries are not distinguishable from zero.

The estimated markup effect is more positive in low-digital than in high-digital industries. The subgroup test below confirms a significant difference between these effects, but the negative high-digital estimate is not individually significant. The results therefore support a difference in markup effects across industry groups without establishing markup compression in the high-digital group or identifying a particular competition mechanism.

The economic magnitudes are modest. A 0.10-point increase in lagged digitalization has an estimated effect of approximately 0.069% on next-period markup in the overall IPC model, conditional on lagged markup. This is smaller than the corresponding overall productivity effects. The unconditional annual mean trend in Figure 1 is a different object and also reflects other shocks and changes in sample composition.

Box: 25th–75th percentiles; central line: median; dashed diamonds: mean

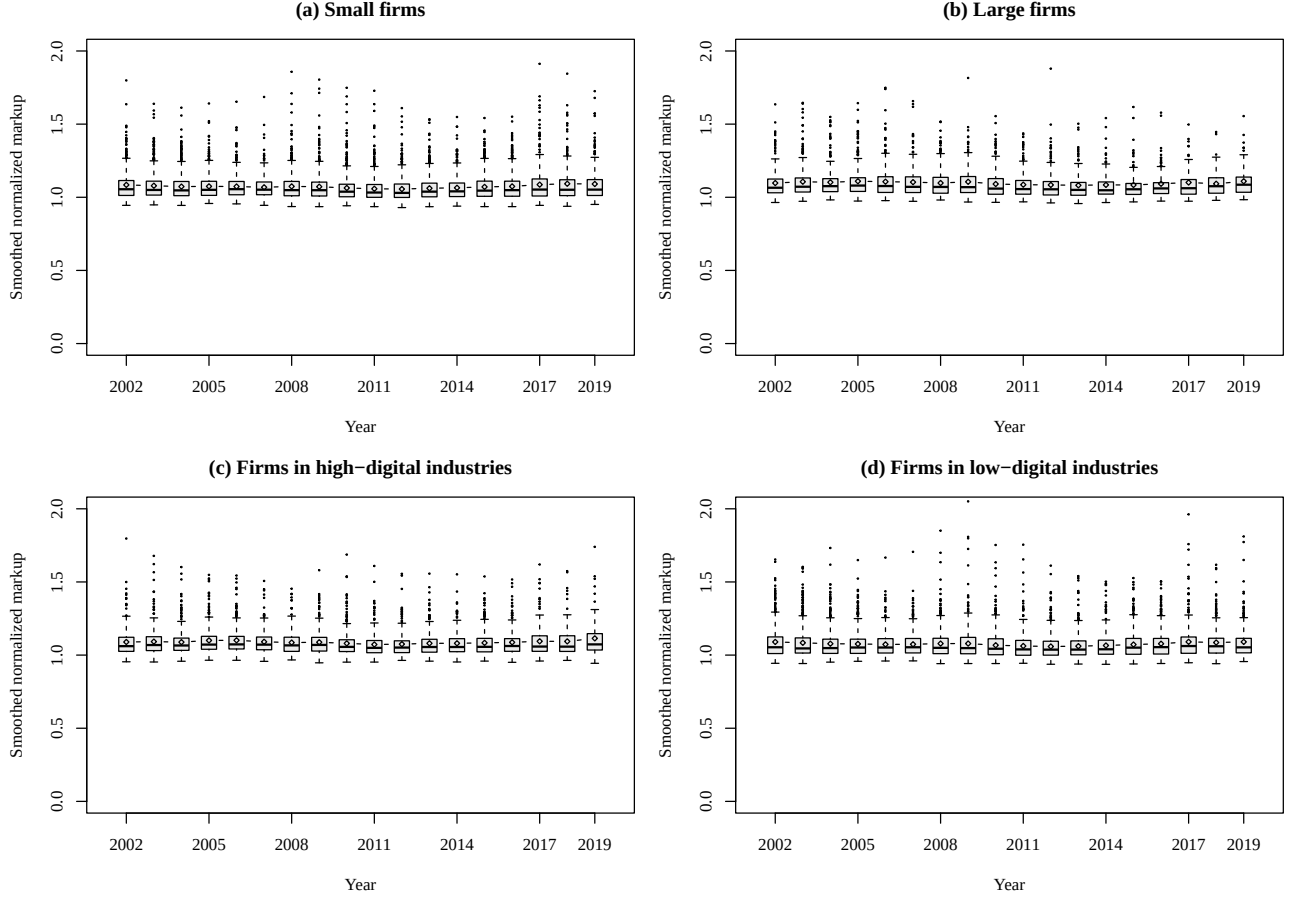


Figure 1: Annual markup distributions by firm size and industry digital intensity

Notes: Unweighted annual distributions of $\exp(\widehat{m}_{it}^{\mu})$, the smoothed markup relative to the unidentified scale θ , from each subgroup's composite-index IPC estimation. Boxes span the 25th–75th percentiles; the central line is the median. Whiskers extend to the most extreme observations within 1.5 interquartile ranges of the box; dots show observations beyond the whiskers. Dashed lines with diamonds show annual means. All panels use the same vertical scale, zoomed to 0–2. All observations enter the boxes and annual means. A full-range version is supplied with the replication figures. Small firms have fewer than 200 employees; large firms have at least 200. The digital groups follow the manuscript's industry classification, not firms' own digitalization scores. The panels contain 9,266, 3,415, 4,523, 8,278 firm-years, respectively. Changes in panel composition can affect annual distributions; boxes are not confidence intervals.

Figure 1 shows annual distributions and mean smoothed normalized markups. For small firms, the annual mean moves from 1.085 in 2002 to its sample-period minimum of 1.056 in 2012. It reaches 1.091 in 2019. The endpoint change is 0.6%. For large firms, the annual mean moves from 1.097 in 2002 to its sample-period minimum of 1.082 in 2013. It reaches 1.110 in 2019. The endpoint change is 1.1%. Technology and markup scale differ across groups. The plotted levels therefore do not rank absolute markups.

For high-digital industries, the annual mean moves from 1.089 in 2002 to its sample-period minimum of 1.072 in 2011. It reaches 1.114 in 2019. The endpoint change is 2.3%. For low-digital industries, the annual mean moves from 1.090 in 2002 to its sample-period minimum of 1.059 in 2012. It reaches 1.090 in 2019. The endpoint change is 0.0%. The groups classify firms by industry digital intensity. Sample composition and technology estimates also affect the comparison.

Calligaris et al. (2018) report rising average markups across 26 countries in 2001–2014 and a widening gap in favor of digitally intensive sectors. Over our overlapping 2002–2014 period, the high-digital mean moves from 1.089 to 1.081, and the low-digital mean from 1.090 to 1.066. The point-normalized series and changing samples describe within-group evolution; differences in period, industry classification, identification, and normalization preclude a formal comparison of absolute markup levels with their cross-country results.

6.2.3 Heterogeneity in the effects of digitalization under IPC

Table 8 reports the corresponding subgroup comparisons for the IPC effects in Table 7.

Under IPC, the small–large labor-augmenting difference is 0.061. The small-firm effect is positive and significant, while the large-firm effect is not statistically distinguishable from zero. The high–low industry difference is 0.012 and is not distinguishable from zero. The estimates therefore provide strong evidence of a firm-size difference, while the industry contrast remains imprecise under IPC.

Under IPC, the Hicks-neutral differences are -0.012 for small minus large firms and -0.030 for high minus low digital industries. Neither difference is significant at ten percent. The small-, large-, and low-digital effects are individually significant, whereas the high-digital estimate is not; these differences in individual significance do not establish differences between groups.

The IPC high-minus-low industry markup difference is -0.018 . Its bootstrap standard error is 0.006. The paired test gives $p = 0.002$. The small-minus-large firm difference is 0.010 with a bootstrap standard error of 0.008. It is not distinguishable from zero, with $p = 0.231$. These tests compare groups. They do not establish a markup reduction within either group.

Table 8: Differences in digitalization effects under IPC

Outcome	Small	Large	Small–large	High	Low	High–low
Labor-augmenting productivity	0.059*** (0.008)	-0.002 (0.011)	0.061*** (0.015)	0.062*** (0.017)	0.050*** (0.010)	0.012 (0.020)
Hicks-neutral productivity	0.042*** (0.010)	0.054** (0.026)	-0.012 (0.028)	0.019 (0.017)	0.049*** (0.010)	-0.030 (0.020)
Log markup	0.013** (0.005)	0.003 (0.007)	0.010 (0.008)	-0.007 (0.004)	0.012*** (0.004)	-0.018 *** (0.006)

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 attempts re-estimating all stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference conditions on preprocessing, the fixed instrument sets and the selected solution branch. Differences use a joint union-firm bootstrap, preserving covariance when firms appear in both groups. Industry digital intensity does not classify individual firms by their own adoption scores. Finite markup difference draws: 499 for size and 499 for industry.

6.2.4 Effects of Digitalization Dimensions on Productivity and Markups

Table 9 reports IPC component estimates for labor-augmenting productivity. Estimates for IT technologies are 0.018 overall and 0.016 for small firms. Estimates for IT technologies are 0.013 in low-digital industries and 0.021 in high-digital industries. Estimates for digital human capital are 0.022 overall and 0.025 for small firms. They are 0.026 in low-digital industries and 0.024 in high-digital industries. Estimates for

automation are 0.011 overall and 0.019 for small firms. The estimated effect of automation is 0.015 in high-digital industries. The estimated effect of stakeholder-facing digitalization is 0.011 overall.

The IPC Hicks-neutral results differ across components. The IT technologies estimates cannot be distinguished from zero in any sample. Estimates for digital human capital are 0.009 overall and 0.010 in low-digital industries. The estimated effect of digital human capital is 0.011 for small firms. Estimates for automation are 0.017 overall and 0.017 for small firms. The effect is 0.022 in low-digital industries. The estimated effect of stakeholder-facing digitalization is 0.013 for small firms.

Automation is the only component with evidence of a markup response. Estimates are 0.009 overall, 0.012 for small firms, and 0.013 in low-digital industries. The other three components cannot be distinguished from zero.

Overall, the IPC component estimates distinguish labor-augmenting, Hicks-neutral, and markup channels. The preceding paragraphs describe component effects by sample, and Table 9 reports all estimates and their bootstrap precision. These results condition on the lagged-capital instrument specification; differences in individual significance do not establish differences between groups.

Table 9: Digitalization components and their effects on productivity and markups under IPC

Component	Overall	Small	Large	Low	High
<i>Labor-augmenting productivity</i>					
IT technologies	0.018*** (0.004)	0.016*** (0.006)	−0.004 (0.019)	0.013** (0.006)	0.021** (0.009)
Digital human capital	0.022*** (0.004)	0.025*** (0.006)	0.012 (0.008)	0.026*** (0.005)	0.024*** (0.009)
Automation	0.011*** (0.004)	0.019*** (0.005)	−0.004 (0.027)	0.008 (0.006)	0.015* (0.009)
Stakeholder-facing digitalization	0.011** (0.006)	0.009 (0.007)	0.004 (0.034)	0.011 (0.007)	0.004 (0.010)
<i>Hicks-neutral productivity</i>					
IT technologies	0.003 (0.004)	0.002 (0.005)	0.008 (0.017)	0.004 (0.006)	0.005 (0.008)
Digital human capital	0.009** (0.004)	0.011* (0.006)	0.015 (0.014)	0.010** (0.005)	0.000 (0.009)
Automation	0.017*** (0.004)	0.017*** (0.006)	0.016 (0.020)	0.022*** (0.005)	0.007 (0.007)
Stakeholder-facing digitalization	0.010 (0.007)	0.013* (0.008)	0.002 (0.022)	0.008 (0.009)	0.011 (0.011)
<i>Log-markup transition</i>					
IT technologies	−0.003 (0.003)	−0.005 (0.003)	0.009 (0.005)	−0.003 (0.004)	0.000 (0.004)
Digital human capital	0.002 (0.002)	0.005 (0.004)	0.002 (0.004)	0.003 (0.003)	−0.004 (0.003)
Automation	0.009*** (0.003)	0.012*** (0.003)	0.003 (0.005)	0.013*** (0.004)	0.004 (0.005)
Stakeholder-facing digitalization	−0.003 (0.004)	0.000 (0.005)	−0.008 (0.006)	−0.002 (0.005)	−0.005 (0.006)
Observations	12,161	8,487	3,085	7,607	4,126
Firms	1,717	1,250	424	1,068	617

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 attempts re-estimating all stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference conditions on preprocessing, the fixed instrument sets and the selected solution branch. Four component indices enter each transition jointly. Finite productivity draws in column order: 499, 499, 499, 499, 499. Finite markup draws: 499, 499, 497, 499, 499.

6.2.5 Summary of results

Taken together, digitalization has positive overall labor-augmenting and Hicks-neutral effects under both models, alongside a smaller markup effect under IPC. The overall labor-augmenting estimates are 0.037 under PC and 0.055 under IPC; the corresponding Hicks-neutral estimates are 0.062 and 0.043. A 0.10-point increase in lagged digitalization corresponds to approximately 0.37% and 0.62% higher next-period labor-augmenting and Hicks-neutral productivity under PC, and 0.55% and 0.43% under IPC.

The composite labor-augmenting effect is significant among small firms under both models and insignificant among large firms. The small–large differences are 0.056 under PC and 0.061 under IPC, both significant at one percent. The high–low industry labor difference is significant under PC and insignificant under IPC. The PC small–large Hicks-neutral difference is significant at five percent, while neither IPC Hicks-neutral subgroup difference is significant at ten percent. Composite Hicks-neutral effects remain positive in every sample, although the high-digital IPC estimate is imprecise.

6.2.6 Separating transitory shocks from Hicks-neutral productivity

Table 10 reports the state-space decomposition. The overall transitory-shock variance is 0.0048. Its bootstrap standard error is 0.0005. Smoothing blackuces the Hicks-neutral signal standard deviation from 0.1618 to 0.1419. These standard deviations describe dispersion across firm-years. The values beneath them in the table are bootstrap standard errors. The identity $H - \widehat{\omega}_N = \widetilde{RR} - \widehat{m}^\mu = \widehat{\eta}$ holds for every observation. Only feasible covariance draws contribute to the smoothed statistics. The smoothed series depend on the state-space assumptions.

Table 10: Transitory-shock extraction under imperfect competition

Outcome	Overall	Small	Large	Low	High
$\hat{\sigma}_\eta^2$	0.0048 (0.0005)	0.0058 (0.0006)	0.0026 (0.0004)	0.0044 (0.0005)	0.0060 (0.0007)
Feasible signal-extraction draws <i>Firm-year means</i>	499	499	499	499	469
Labor-augmenting productivity	0.3173 (0.0351)	0.2443 (0.0326)	-0.0869 (0.1101)	0.3468 (0.0659)	0.2305 (0.1456)
Hicks-neutral signal	0.0647 (0.0059)	0.0621 (0.0068)	0.0683 (0.0205)	0.0708 (0.0120)	0.0605 (0.0167)
Smoothed Hicks-neutral productivity	0.0643 (0.0059)	0.0619 (0.0068)	0.0677 (0.0205)	0.0704 (0.0120)	0.0599 (0.0166)
Pblackicted transitory shock	0.0004 (0.0001)	0.0003 (0.0002)	0.0006 (0.0002)	0.0004 (0.0001)	0.0005 (0.0004)
Smoothed normalized log markup	0.0666 (0.0098)	0.0658 (0.0167)	0.0878 (0.0704)	0.0681 (0.0145)	0.0806 (0.0728)
<i>Firm-year standard deviations</i>					
Labor-augmenting productivity	0.4686 (0.0136)	0.4410 (0.0146)	0.3697 (0.0797)	0.5069 (0.0346)	0.3739 (0.0996)
Hicks-neutral signal	0.1618 (0.0033)	0.1635 (0.0051)	0.1551 (0.0791)	0.1648 (0.0348)	0.1535 (0.0631)
Smoothed Hicks-neutral productivity	0.1419 (0.0035)	0.1407 (0.0056)	0.1379 (0.0805)	0.1475 (0.0356)	0.1282 (0.0677)
Pblackicted transitory shock	0.0519 (0.0044)	0.0601 (0.0058)	0.0348 (0.0049)	0.0491 (0.0044)	0.0673 (0.0073)
Smoothed normalized log markup	0.0862 (0.0037)	0.0845 (0.0051)	0.0946 (0.0079)	0.0864 (0.0034)	0.0808 (0.0042)

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 attempts re-estimating all stages. Stars use two-sided, un-adjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference conditions on preprocessing, the fixed instrument sets and the selected solution branch. The reported $\hat{\sigma}_\eta^2$ is the common transitory-shock variance used in both measurement equations of the Kalman filter and smoother, estimated from the markup equation's lag-one residual covariance.

7 Diagnostics and Robustness

We perform robustness checks in this section.

7.1 Technology and persistence estimates

The diagnostics and robustness checks reported in Tables 11 and 12 assess the fitted technology, instrument restrictions, and precision of the estimates. They help qualify the interpretation of the results rather than independently validating the model assumptions.

7.1.1 perfect-competition

Under PC, median returns to scale range from 0.949 to 0.987 across samples. Mean capital elasticity is positive in every sample, ranging from 0.078 to 0.113; no retained firm-year has a negative estimated capital derivative. The sign of β_K alone does not determine monotonicity because $\varepsilon_{it}^K = \beta_K + \beta_{KK}k_{it}$.

The PC persistence estimates range from 0.796 to 0.903 across the two productivity processes and samples. The estimates indicate persistence in both productivity processes.

Table 11: Technology, persistence and estimation diagnostics under PC

Parameter	Overall	Small	Large	Low	High
ρ_L	0.891*** (0.005)	0.884*** (0.005)	0.903*** (0.011)	0.901*** (0.005)	0.862*** (0.010)
ρ_N	0.840*** (0.011)	0.853*** (0.013)	0.796*** (0.027)	0.837*** (0.014)	0.838*** (0.021)
ν	0.872*** (0.002)	0.871*** (0.002)	0.876*** (0.004)	0.872*** (0.002)	0.873*** (0.003)
δ	-0.129*** (0.003)	-0.139*** (0.004)	-0.111*** (0.009)	-0.123*** (0.004)	-0.137*** (0.006)
β_K	-0.129*** (0.041)	-0.159*** (0.062)	-0.058 (0.194)	-0.141*** (0.053)	-0.072 (0.066)
β_{KK}	0.0156*** (0.0029)	0.0174*** (0.0047)	0.0099 (0.0110)	0.0168*** (0.0038)	0.0102** (0.0045)
Median returns to scale	0.969	0.949	0.987	0.977	0.950
Mean capital elasticity	0.098	0.078	0.113	0.106	0.078
Share with negative capital elasticity	0.000	0.000	0.000	0.000	0.000
Observations	17,087	12,246	4,343	10,610	5,883
Firms	2,213	1,689	546	1,374	789

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 attempts re-estimating all stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference conditions on preprocessing, the fixed instrument sets and the selected solution branch. Step 2 is exactly identified; Step 3 is nonlinear least squares. The selected branch favors positive variable-input effects and then the lowest Step-3 residual sum of squares; bootstrap draws match that branch.

7.1.2 IPC

The IPC Hansen reference p -values range from 0.123 to 1.000. They do not reject the overidentifying restrictions at conventional levels; these reference tests alone do not establish instrument validity.

Mean IPC returns to scale range from 0.971 to 1.003. The large-firm estimate is close to one and slightly above it, while the other point estimates are below one. Mean capital elasticity is positive in every IPC sample, ranging from 0.031 to 0.042. In the large-firm sample, 10.2% of firm-years nevertheless have negative capital elasticity; the other composite samples have none.

IPC labor-augmenting persistence ranges from 0.906 to 0.942, Hicks-neutral persistence from 0.510 to 0.819, and markup persistence from 0.836 to 0.892. The large-firm Hicks-neutral persistence estimate of 0.510 is lower than those in the other samples.

Table 12: Technology, persistence and estimation diagnostics under IPC

Parameter	Overall	Small	Large	Low	High
ρ_L	0.934*** (0.005)	0.919*** (0.006)	0.940*** (0.014)	0.942*** (0.007)	0.906*** (0.016)
ρ_N	0.777*** (0.029)	0.817*** (0.034)	0.510*** (0.172)	0.788*** (0.033)	0.819*** (0.097)
ρ_μ	0.846*** (0.019)	0.859*** (0.024)	0.846*** (0.038)	0.836*** (0.022)	0.892*** (0.032)
ν	0.939*** (0.009)	0.938*** (0.016)	0.961*** (0.092)	0.941*** (0.014)	0.951*** (0.078)
δ	-0.206*** (0.006)	-0.209*** (0.008)	-0.176*** (0.043)	-0.203** (0.089)	-0.198 (0.217)
β_K	-0.016 (0.030)	-0.022 (0.078)	0.554 (1.640)	-0.021 (0.242)	0.017 (0.231)
β_{KK}	0.0038* (0.0021)	0.0041 (0.0060)	-0.0297 (0.0923)	0.0039 (0.0165)	0.0010 (0.0170)
Mean returns to scale	0.978	0.971	1.003	0.978	0.982
Mean capital elasticity	0.039	0.034	0.042	0.037	0.031
Share with negative capital elasticity	0.000	0.000	0.102	0.000	0.000
Hansen reference p	0.686	0.539	1.000	0.522	0.123
Mean $\hat{\nu}/R$	1.078 (0.011)	1.077 (0.019)	1.100 (0.106)	1.078 (0.016)	1.092 (0.089)
Observations	13,291	9,266	3,415	8,278	4,523
Firms	1,816	1,324	457	1,120	656

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 attempts re-estimating all stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7.2 Alternative materials-demand proxy

The alternative specification replaces the main Hicks-neutral block with a lagged materials-demand proxy adjusted by a forecast of lagged markup. Appendix B gives its construction and additional exclusion restrictions. Table 13 reports a fresh native-R rerun on the same coded-capital IPC panels.

Table 13: Sensitivity using a markup-adjusted materials proxy

Outcome	Overall	Small	Large	Low	High
Labor-augmenting productivity	0.061*** (0.010)	0.053*** (0.013)	0.006 (0.014)	0.033*** (0.012)	0.055*** (0.013)
Hicks-neutral productivity	0.039*** (0.013)	0.014 (0.010)	0.005 (0.009)	0.016** (0.008)	0.025 (0.017)
Log-markup transition	0.007** (0.003)	0.013*** (0.005)	0.003 (0.006)	0.012*** (0.004)	-0.007 (0.004)
ν	0.118	0.907	1.004	0.909	0.833
ρ_N	0.987	0.950	0.921	0.947	0.963
Draws at $\nu = 0.05$	168	10	3	0	5
Draws at $\rho_N = 0.995$	2	29	0	0	6
Observations	13,291	9,266	3,415	8,278	4,523
Firms	1,816	1,324	457	1,120	656
Successful bootstrap draws	499	499	499	499	499

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 attempts re-estimating all stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference conditions on preprocessing, the fixed instrument sets and the selected solution branch. This rerun uses the corrected IPC composite panels but retains the separate labor-first specification in Appendix B. It imposes $0.05 \leq \nu \leq 3$, $-0.95 \leq \rho_N \leq 0.995$, and capital elasticity at least 10^{-6} on the retained support. Capital curvature is estimated in all five groups. Normal-reference stars are not boundary-calibrated. The markup forecast needs additional exclusion restrictions; this is not the MZZ IPC point estimator, nor a comparison changing only the Hicks proxy.

The alternative-proxy specification gives the following results. For the overall sample, the labor-augmenting estimate is 0.061. For small firms, the labor-augmenting estimate is 0.053. For large firms, the labor-augmenting estimate is 0.006. For low-digital industries, the labor-augmenting estimate is 0.033. For high-digital industries, the labor-augmenting estimate is 0.055. The estimate for large firms is not distinguishable from zero. For the overall sample, the Hicks-neutral estimate is 0.039. For small firms, the Hicks-neutral estimate is 0.014. For large firms, the Hicks-neutral estimate is 0.005. For low-digital industries, the Hicks-neutral estimate is 0.016. For high-digital industries, the Hicks-neutral estimate is 0.025. The estimates for small firms, large firms and high-digital industries are not distinguishable from zero. For the overall sample, the markup-transition estimate is 0.007. For small firms, the markup-transition estimate is 0.013. For large firms, the markup-transition estimate is 0.003. For low-digital industries, the markup-transition estimate is 0.012. For high-digital industries, the markup-transition estimate is -0.007. The estimates for large firms and high-digital industries are not distinguishable from zero.

The alternative uses a labor-first sequence, whereas the main IPC specification estimates Hicks-neutral technology first and then labor dynamics conditional on that technology. The overall labor effects are 0.061 in the alternative and 0.055 in the main IPC model. The alternative also imposes parameter bounds and positive capital elasticity throughout its fixed support. Its optimizer may select different branches across samples and bootstrap draws; the estimates in the table are the freely fitted native-R results. This comparison changes the estimation sequence and restrictions as well as the Hicks-neutral proxy.

8 Policy Implications

Our findings provide an empirical rationale for policies that support digital adoption while distinguishing productive gains from changes in market power. Under perfect competition, a 0.10-point increase in the digitalization index is associated with approximately 0.37% higher labor-augmenting productivity and 0.62% higher Hicks-neutral productivity in the following period. Under imperfect competition, the corresponding estimates are 0.55% and 0.43%, alongside a comparatively small markup response of 0.07%. These conditional relationships suggest that digital adoption can improve the efficiency of existing labor and expand output at given effective inputs. Policy evaluations should therefore measure these productivity channels separately from changes in the price–marginal-cost wedge, since improvements in productive capacity and increases in markups have different implications for the distribution of economic gains.

The firm-size results give a specific reason to consider smaller manufacturers when designing adoption support. The labor-augmenting response is significantly larger among firms with fewer than 200 workers than among larger firms under both competitive specifications. Programs that improve access to digital equipment, implementation assistance, and relevant training could therefore help smaller firms strengthen their labor efficiency. This finding does not imply that support for large firms has no productive value: their Hicks-neutral response remains positive and significant. Instead, it suggests that program objectives should reflect the different productivity channels observed across firm sizes. Whether smaller firms should receive greater public support also depends on adoption costs, financing constraints, and the additional

investment generated by that support, which our estimates do not directly identify.

The component estimates further suggest that the composition of adoption should inform policy design. Among small firms, IT technologies, digital human capital, and automation each have positive and significant labor-augmenting responses under imperfect competition. The digital human-capital result provides a reason to evaluate IT training alongside technology acquisition, while the IT and automation results indicate that labor-efficiency gains are associated with several forms of digital investment. Automation also has positive Hicks-neutral and markup responses in the overall sample and in several subgroups. Consequently, support programs should track which technologies firms adopt and which outcomes improve, rather than treating all digital investments as equivalent. These findings motivate evaluating programs that combine skills development with technology access, although they do not directly establish complementarities or determine the optimal allocation of subsidies.

The markup findings make competition conditions relevant to the assessment of digital adoption policies. The estimated markup response is significantly smaller in high-digital-intensity industries than in low-digital-intensity industries. This distinction complements the higher markup levels in digitally intensive industries documented by Calligaris et al. (2018): industries with higher markups need not exhibit a larger response to additional adoption. Competition assessments should therefore consider both existing markup levels and their evolution as firms digitalize. Where adoption is associated with productivity improvements and higher markups, further evidence on marginal costs and consumer prices is needed to establish how much of the productive gain reaches consumers. The negative but individually insignificant estimate in high-digital-intensity industries does not establish that adoption reduces market power.

Finally, the estimates offer quantitative benchmarks for designing and evaluating adoption programs. A 0.10-point increase in the index provides a common scale for reporting the estimated relationships, but it does not represent a fixed expenditure or a particular technology purchase. Because productivity and markups are persistent, evaluations should also examine whether gains accumulate when higher adoption is sustained. Establishing broader policy benefits requires tracking wages, employment, consumer prices, and implementation costs alongside firm productivity and markups. Labor-augmenting productivity alone does not determine wages, employment, or the labor share, while a higher markup alone does not establish a consumer welfare loss. Our estimates identify conditional relationships within the maintained production framework; causal returns to specific interventions require additional evidence. Their policy value lies in identifying promising outcomes and groups to examine, and in providing benchmarks against which intervention-specific results can be assessed.

9 Conclusion

This paper examines digitalization, labor-augmenting and Hicks-neutral productivity, and firm markups under PC and IPC. Overall productivity effects are positive and significant in both models. The composite labor-augmenting response is significantly larger among small firms under both specifications. The PC firm-size Hicks-neutral difference is significant at five percent, whereas the other Hicks-neutral sub-

group differences are imprecise. The overall IPC markup response is small, positive, and significant; it is significantly lower in high-digital industries than in low-digital industries. Component estimates reveal several productive channels, with automation contributing to Hicks-neutral productivity and markups in the overall, small-firm, and low-digital IPC samples. Capital construction follows the coded industry-size initialization and perpetual-inventory recursion. Mean capital elasticity is positive in all main samples, although a positive mean does not ensure a positive derivative for every firm-year.

The digitalization index combines technologies with different productivity effects. Those effects vary by firm size, industry digital intensity, and technology. Separating productivity from markups requires identifying restrictions. Comparing the two models shows how the results depend on those restrictions.

These findings distinguish productivity effects with different implications for the labor share. A labor-augmenting effect alone does not determine that share, which also depends on substitution between inputs and market power. The distinction is relevant to the discussion of labor-complementing technologies in Acemoglu et al. (2026). Our estimates show positive labor-augmenting effects of digital human capital in several samples, while automation has both labor-augmenting and Hicks-neutral effects in some samples. They therefore do not support treating digital human capital as the exclusive labor-augmenting channel or automation as exclusively Hicks-neutral. The results concern firm productivity and markups and do not directly estimate effects on workers or the returns to particular policy interventions.

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Appendix

A Distributions of Elasticities, Productivities and Markups

Table 14 summarizes the PC estimates. In the overall sample, mean output elasticities are 0.098 for capital, 0.235 for labor, and 0.637 for materials. Labor and materials elasticities sum to the estimated variable-input scale ν by construction. Their equal standard deviations follow from these identities rather than providing independent evidence about technology.

The final two columns of Table 14 report recovered log productivity after subtracting each sample's 2001 median. These are overall firm-year distributions relative to a common within-sample reference point. A negative observation is below that reference median. Labor-augmenting productivity has greater dispersion in its own log units than Hicks-neutral productivity, but these components enter output differently, so their dispersions are not directly comparable output contributions.

Table 14: PC elasticities and normalized productivity distributions

Sample	ε_K	ε_L	ε_M	$\tilde{\omega}_L$	$\tilde{\omega}_N$
Overall	0.098 (0.032) [0.073, 0.097, 0.123]	0.235 (0.126) [0.139, 0.214, 0.307]	0.637 (0.126) [0.565, 0.658, 0.732]	0.094 (0.445) [-0.205, 0.101, 0.376]	0.044 (0.194) [-0.061, 0.062, 0.168]
Small firms	0.078 (0.026) [0.059, 0.078, 0.099]	0.254 (0.133) [0.153, 0.235, 0.335]	0.617 (0.133) [0.536, 0.636, 0.718]	0.174 (0.429) [-0.125, 0.177, 0.462]	0.059 (0.186) [-0.049, 0.076, 0.180]
Large firms	0.113 (0.011) [0.105, 0.112, 0.119]	0.187 (0.089) [0.118, 0.181, 0.238]	0.689 (0.089) [0.638, 0.695, 0.757]	0.034 (0.387) [-0.214, 0.016, 0.257]	0.075 (0.170) [-0.025, 0.075, 0.175]
Low-digital industries	0.106 (0.035) [0.078, 0.105, 0.133]	0.225 (0.132) [0.125, 0.200, 0.297]	0.647 (0.132) [0.575, 0.673, 0.747]	0.095 (0.480) [-0.226, 0.088, 0.396]	0.048 (0.207) [-0.061, 0.072, 0.178]
High-digital industries	0.078 (0.020) [0.061, 0.076, 0.094]	0.249 (0.110) [0.169, 0.234, 0.314]	0.624 (0.110) [0.560, 0.640, 0.704]	0.140 (0.383) [-0.114, 0.150, 0.396]	0.044 (0.184) [-0.066, 0.058, 0.167]

Notes: Cells show mean (SD) and [P25, median, P75]. Productivity is centered on the median of the first estimation year in each sample. The materials inversion and output-minus-transitory-shock recovery of Hicks productivity agree to numerical precision after coherent preprocessing.

Tables 15 and 16 report IPC distributions. They follow the format of Table 14. The first reports input elasticities and log productivities. The second reports smoothed normalized markups in logs and levels. Standard deviations in these appendix tables describe dispersion across firm-years. They are not bootstrap standard errors.

Table 15: IPC elasticities and normalized productivity distributions

Sample	ε_K	ε_L	ε_M	$\tilde{\omega}_L$	$\tilde{\omega}_N$
Overall	0.039 (0.008) [0.033, 0.039, 0.045]	0.249 (0.134) [0.148, 0.227, 0.325]	0.689 (0.134) [0.613, 0.712, 0.791]	0.122 (0.469) [-0.200, 0.111, 0.425]	0.067 (0.142) [-0.021, 0.057, 0.149]
Small firms	0.034 (0.006) [0.029, 0.034, 0.038]	0.271 (0.142) [0.164, 0.249, 0.355]	0.667 (0.142) [0.582, 0.689, 0.774]	0.157 (0.441) [-0.146, 0.142, 0.435]	0.067 (0.141) [-0.020, 0.056, 0.146]
Large firms	0.042 (0.032) [0.023, 0.045, 0.065]	0.203 (0.097) [0.129, 0.197, 0.259]	0.758 (0.097) [0.702, 0.764, 0.832]	0.016 (0.370) [-0.239, 0.013, 0.256]	0.064 (0.138) [-0.029, 0.051, 0.143]
Low-digital industries	0.037 (0.008) [0.031, 0.037, 0.043]	0.238 (0.140) [0.132, 0.212, 0.315]	0.702 (0.140) [0.625, 0.729, 0.809]	0.138 (0.507) [-0.218, 0.116, 0.476]	0.068 (0.148) [-0.024, 0.055, 0.151]
High-digital industries	0.031 (0.002) [0.030, 0.031, 0.033]	0.269 (0.118) [0.183, 0.252, 0.338]	0.682 (0.118) [0.612, 0.699, 0.768]	0.112 (0.374) [-0.137, 0.125, 0.365]	0.072 (0.128) [-0.008, 0.065, 0.149]

Notes: Cells show mean (SD) and [P25, median, P75] across unweighted firm-year observations; parentheses contain distributional standard deviations. Quartiles use the same median-unbiased convention as Table 14. The overall, small, large, low-digital, and high-digital samples contain 13,291, 9,266, 3,415, 8,278, 4,523 observations, respectively. Estimates use the composite digitalization index and lagged-capital instruments.

Table 16: IPC normalized markup distributions

Sample	$\hat{m}^\mu$	$\exp(\hat{m}^\mu)$
Overall	0.067 (0.086) [0.008, 0.046, 0.101]	1.073 (0.104) [1.008, 1.047, 1.107]
Small firms	0.066 (0.085) [0.008, 0.045, 0.100]	1.072 (0.100) [1.008, 1.046, 1.105]
Large firms	0.088 (0.095) [0.027, 0.063, 0.118]	1.097 (0.120) [1.028, 1.065, 1.126]
Low-digital industries	0.068 (0.086) [0.008, 0.047, 0.105]	1.075 (0.101) [1.008, 1.048, 1.111]
High-digital industries	0.081 (0.081) [0.027, 0.061, 0.111]	1.088 (0.095) [1.027, 1.063, 1.118]

Notes: Cells show mean (SD) and [P25, median, P75] across unweighted firm-year observations; parentheses contain distributional standard deviations. Quartiles use the same median-unbiased convention as Table 14. The overall, small, large, low-digital, and high-digital samples contain 13,291, 9,266, 3,415, 8,278, 4,523 observations, respectively. Estimates use the composite digitalization index and lagged-capital instruments. Here $\hat{m}_{it}^\mu = \ln \mu_{it} - \ln \theta$. Columns report smoothed normalized log markup and its exponential, the markup relative to the common scale θ .

B Robustness Check Using Lagged Materials Demand as the Hicks-Neutral Proxy

The robustness check uses an alternative proxy for lagged Hicks-neutral productivity within the IPC model. It draws on the materials-demand inversion idea, but the markup-forecast correction is an additional construction required here and is not part of the PC estimator. The Stage 1 labor-productivity moments, technology, and markup definition are maintained.

B.1 Materials-demand inversion under IPC

Malikov et al. (2024) extend their method to IPC by using materials demand as a proxy for lagged unobserved Hicks-neutral productivity. We adopt the same proxy approach in this robustness check.

Let $w_{it}^M \equiv \ln W_{it}^M$, $p_{it} \equiv \ln P_{it}$, and $\mathbf{m}_{it} \equiv \ln \mu_{it}$. The materials first-order condition can be written in logs as

$$w_{it}^M - p_{it} = \ln \theta - \mathbf{m}_{it} + \ln \varepsilon_{it}^M + \ln F(K_{it}, L_{it}^*, M_{it}) + \omega_{N,it} - m_{it}, \quad (71)$$

where $\varepsilon_{it}^M = \nu SC_{it}^M$ from the share identities. The constant $\ln \theta$ is retained separately from κ_0 ; it is carried explicitly here to keep the derivation transparent. η_{it} does not enter this condition because it affects measured output and revenue, not the productivity state observed by the firm. From equation (18), $\varepsilon_{M,it}^* = \nu SC_{it}^M$, where $SC_{it}^M = 1 - SC_{it}^L$. Define the scale-free materials-demand index

$$\mathcal{M}_{it}^*\left(\nu, \frac{\nu}{\delta}\right) \equiv (w_{it}^M - p_{it}) - \ln SC_{it}^M + (1 - \nu)m_{it} + \left(\frac{\nu^2}{2\delta}\right)(SC_{it}^L)^2. \quad (72)$$

Substitution of the reduced restricted translog technology in equation (19) into equation (71) gives

$$\mathcal{M}_{it}^*\left(\nu, \frac{\nu}{\delta}\right) = \kappa_0 + \beta_K k_{it} + \frac{1}{2}\beta_{KK} k_{it}^2 + \omega_{N,it} - \mathbf{m}_{it} + \ln \nu + \ln \theta. \quad (73)$$

Thus, under IPC, raw materials demand identifies Hicks-neutral productivity net of the log markup. Treating $m_{i,t-1}$ one-for-one as $\omega_{N,i,t-1}$ would impose an incorrect scale and load persistent markup variation into productivity.

The required markup correction follows from the maintained markup block. Because $R_{it} = \theta \nu \mu_{it}^{-1} e^{-\eta_{it}}$, define the scale-free observed log-markup measure

$$b_{it} \equiv -\ln R_{it} = \mathbf{m}_{it} + \eta_{it} - \ln \nu - \ln \theta. \quad (74)$$

Adding a constant to the dependent variable of equation (61) changes only its intercept. Consequently, b_{it} follows the same transition with intercept $\pi_b = \pi_\mu - (1 - \rho_\mu) \ln(\theta \nu)$ and unchanged slopes (ρ_μ, γ_μ) . Because $\ln \mu_{i,t-1}$ is latent and the observed $b_{i,t-1}$ contains $\eta_{i,t-1}$, it cannot serve as its own instrument. We estimate the observed-signal transition using a constant, $b_{i,t-2}$, and predetermined digitalization as instruments, and then form the one-step forecast

$$\widehat{b}_{i,t-1|t-2} = \widehat{\pi}_b + \widehat{\rho}_\mu(-\ln R_{i,t-2}) + \widehat{\gamma}_\mu D_{i,t-2}. \quad (75)$$

The lagged materials-demand proxy for the location-normalized Hicks-neutral state is

$$\mathcal{H}_{i,t-1}^M\left(\boldsymbol{\Theta}, \frac{\nu}{\delta}\right) \equiv \mathcal{M}_{i,t-1}^*\left(\nu, \frac{\nu}{\delta}\right) - \beta_K k_{i,t-1} - \frac{1}{2}\beta_{KK} k_{i,t-1}^2 + \widehat{b}_{i,t-1|t-2}. \quad (76)$$

Under a correctly specified markup transition, $\mathcal{H}_{i,t-1}^M = \kappa_0 + \omega_{N,i,t-1} + e_{i,t-1}^M$, where $e_{i,t-1}^M$ combines the lagged transitory output shock $\eta_{i,t-1}$ and the negative of the markup-forecast error. The constant κ_0 is absorbed by the dynamic intercept below.

B.2 Alternative sequential nonlinear-GMM block

Substituting equation (76) for the lagged state in equation (46) gives the robustness-check residual

$$u_{n,it}^M(\Theta) = y_{it} - \pi_0 - \beta_K k_{it} - \frac{1}{2} \beta_{KK} k_{it}^2 - \nu m_{it} + \frac{\nu}{2} \left(\frac{\nu}{\delta} \right) (SC_{it}^L)^2 - \rho_N \mathcal{H}_{i,t-1}^M \left(\Theta, \left(\frac{\nu}{\delta} \right) \right) - \gamma_N D_{i,t-1}, \quad \pi_0 = \pi_N + (1 - \rho_N) \kappa_0. \quad (77)$$

As in the main IPC specification, the location of the recovered Hicks-neutral composite is fixed by setting its first-estimation-year median to zero. This normalization does not recover the primitive production-function parameters. Its composite disturbance is $\xi_{N,it} + \eta_{it} - \rho_N e_{i,t-1}^M$. Relative to the benchmark dynamic production equation, lagged observed output no longer appears directly. The generated markup-forecast error enters instead. Conditional on the Stage 1 estimate of ν/δ , estimate $(\pi_0, \beta_K, \beta_{KK}, \nu, \rho_N, \gamma_N)'$ by two-step nonlinear GMM using

$$\mathbb{E}[\mathbf{Z}_{n,it}^M u_{n,it}^M(\Theta_0)] = \mathbf{0}, \quad (78)$$

where the baseline instrument vector is

$$\mathbf{Z}_{n,it}^M = (1, D_{i,t-1}, m_{i,t-1}, k_{it}, k_{it}^2, p_{i,t-1}^M - p_{i,t-1})'. \quad (79)$$

Capital dated t is predetermined by assumption. The lagged variable inputs and relative input price instrument the current static choices and the materials-demand state. These six moments exactly identify the six Hicks-neutral/technology parameters conditional on the Stage 1 estimate of ν/δ .

The estimation remains sequential. First, we estimate the Stage 1 labor-productivity block using equations (57)–(59). Second, we estimate the scale-free markup transition in equation (61) and construct the markup forecast in equation (75). Third, we estimate the Hicks-neutral productivity block in equation (78) by nonlinear GMM and re-anchor the markup intercept once ν is identified. Each firm-block bootstrap draw repeats the full estimation sequence. The cleaning rules and five estimation samples are the same as in the benchmark specification. The output and materials prices introduced in this robustness check are winsorized at the 2.5th and 97.5th percentiles.

B.3 Identification conditions

The robustness check requires an interior materials choice, $\nu SC_{it}^M > 0$, correct measurement of relative materials prices, and price taking in the materials market. The fitted markup forecast must remove the conditional markup component of materials demand, and its forecast error must be orthogonal to the instruments in equation (79). Lagged digitalization must not shift materials demand directly after conditioning on Hicks-neutral productivity, the markup process, and the included state variables. These are additional assumptions rather than consequences of using a lag. If they fail, γ_N combines technological adoption with an omitted input-demand or markup channel.

C Signal Extraction for the Markup Innovation

This appendix develops a signal-extraction procedure that separates the latent log markup from the transitory output shock at the firm-year level. The procedure is an extension of the main IPC model. It is not used to construct the baseline estimates in Section 6. It requires additional restrictions on the second moments of the markup innovation and the transitory output shock.

C.1 State and measurement equations

Define

$$\widetilde{RR}_{it} \equiv \ln \widehat{\nu} - \ln R_{it}.$$

Ignoring first-step estimation error in $\widehat{\nu}$ for exposition, equations (61) and (62) imply the state-space representation

$$\text{state:} \quad \widetilde{m}_{it}^{\mu} = \widetilde{\pi}_{\mu} + \rho_{\mu} \widetilde{m}_{i,t-1}^{\mu} + \gamma_{\mu} D_{i,t-1} + \xi_{\mu,it}, \quad (80)$$

$$\text{measurement:} \quad \widetilde{RR}_{it} = \widetilde{m}_{it}^{\mu} + \eta_{it}, \quad (81)$$

where $\widetilde{m}_{it}^{\mu} \equiv \ln \mu_{it} - \ln \theta$ is the log markup normalized relative to $\ln \theta$. Because θ is constant, this normalization changes only the transition intercept: $\widetilde{\pi}_{\mu} = \pi_{\mu} - (1 - \rho_{\mu}) \ln \theta$. The persistence parameter, the digitalization effect, and the innovation are unchanged. Hence the Stage 3 composite disturbance is

$$u_{\mu,it} = \xi_{\mu,it} + \eta_{it} - \rho_{\mu} \eta_{i,t-1}. \quad (82)$$

Identification of the two components comes instead from their distinct serial covariance pattern.

C.2 Variance identification

Let

$$\sigma_{\xi}^2 \equiv \text{Var}(\xi_{\mu,it}), \quad \sigma_{\eta}^2 \equiv \text{Var}(\eta_{it}),$$

with both variances constant within an estimation sample.

Assumption B.1 (Second-moment restrictions). The sequences $\{\xi_{\mu,it}\}$ and $\{\eta_{it}\}$ have mean zero and are serially uncorrelated. They are mutually orthogonal at all leads and lags used below. Their variances are finite and constant within the estimation sample. In addition,

$$0 < |\rho_{\mu}| < 1.$$

The restriction $\rho_{\mu} \neq 0$ is needed to identify σ_{η}^2 from the first-order autocovariance. Values of $|\rho_{\mu}|$ close to zero imply weak variance separation in finite samples.

Proposition B.1 (Variance separation). Under Assumption B.1 and the Stage 3 transition,

$$\text{Var}(u_{\mu,it}) = \sigma_\xi^2 + (1 + \rho_\mu^2)\sigma_\eta^2, \quad (83)$$

$$\text{Cov}(u_{\mu,it}, u_{\mu,i,t-1}) = -\rho_\mu\sigma_\eta^2, \quad (84)$$

$$\text{Cov}(u_{\mu,it}, u_{\mu,i,t-h}) = 0, \quad h \geq 2. \quad (85)$$

Proof. Using equation (82),

$$u_{\mu,it} = \xi_{\mu,it} + \eta_{it} - \rho_\mu\eta_{i,t-1}.$$

The variance result follows because the three terms are pairwise orthogonal under Assumption B.1. For the lag-one covariance, the only nonzero cross-product is generated by $\eta_{i,t-1}$, which enters $u_{\mu,it}$ with effect $-\rho_\mu$ and $u_{\mu,i,t-1}$ with effect one. This gives

$$\text{Cov}(u_{\mu,it}, u_{\mu,i,t-1}) = -\rho_\mu\sigma_\eta^2.$$

At lags $h \geq 2$, the two composite disturbances contain no common transitory-shock term, and all remaining cross-products are zero by Assumption B.1. This gives equation (85). $\square$

Because $\rho_\mu \neq 0$, equations (83) and (84) identify the two disturbance variances:

$$\sigma_\eta^2 = -\frac{\text{Cov}(u_{\mu,it}, u_{\mu,i,t-1})}{\rho_\mu}, \quad (86)$$

$$\sigma_\xi^2 = \text{Var}(u_{\mu,it}) - (1 + \rho_\mu^2)\sigma_\eta^2. \quad (87)$$

The sample counterparts are

$$\widehat{\sigma}_\eta^2 = -\frac{\widehat{\text{Cov}}(\hat{u}_{\mu,it}, \hat{u}_{\mu,i,t-1})}{\hat{\rho}_\mu}, \quad (88)$$

$$\widehat{\sigma}_\xi^2 = \widehat{\text{Var}}(\hat{u}_{\mu,it}) - (1 + \hat{\rho}_\mu^2)\widehat{\sigma}_\eta^2. \quad (89)$$

Equation (85) provides overidentifying specification checks. In particular, the sample autocovariances of $\hat{u}_{\mu,it}$ at lags two and above should be close to zero. A sizable higher-order autocovariance indicates misspecification of the AR(1) markup process, serial correlation in η_{it} , or both.

The variance estimates must also satisfy

$$\widehat{\sigma}_\eta^2 \geq 0, \quad \widehat{\sigma}_\xi^2 \geq 0.$$

Negative sample estimates indicate that the maintained second-moment restrictions are not supported by that sample. In that case, the signal-extraction step should not be imposed mechanically.

C.3 Kalman filtering and smoothing

Given

$$(\widehat{\pi}_\mu, \hat{\rho}_\mu, \hat{\gamma}_\mu, \widehat{\sigma}_\eta^2, \widehat{\sigma}_\xi^2),$$

equations (80)–(81) form a linear state-space model. The filtering and smoothing are carried out separately for each firm and, in the baseline implementation, separately for each consecutive-year spell. This avoids treating a multi-year gap as a one-period transition. If missing calendar years are retained instead, the transition matrix and accumulated state-innovation variance must be adjusted for the length of the gap.

Let $x_{t|s}$ denote the linear projection of $\tilde{m}_{it}^\mu$ using information through period s , and let $P_{t|s}$ denote its mean-squared prediction error. We use diffuse initialization for the first observation of each consecutive spell. The standard Kalman recursions shown below apply after this initialization step:

$$x_{t|t-1} = \widehat{\pi}_\mu + \hat{\rho}_\mu x_{t-1|t-1} + \hat{\gamma}_\mu D_{i,t-1}, \quad P_{t|t-1} = \hat{\rho}_\mu^2 P_{t-1|t-1} + \widehat{\sigma}_\xi^2, \quad (90)$$

$$K_t = \frac{P_{t|t-1}}{P_{t|t-1} + \widehat{\sigma}_\eta^2}, \quad x_{t|t} = x_{t|t-1} + K_t (\widetilde{RR}_{it} - x_{t|t-1}), \quad (91)$$

$$P_{t|t} = (1 - K_t) P_{t|t-1}. \quad (92)$$

Because the sample is historical rather than a real-time forecasting exercise, we use the Rauch–Tung–Striebel smoother after the forward pass. For $t = T - 1, \dots, 1$,

$$J_t = \frac{P_{t|t} \hat{\rho}_\mu}{P_{t+1|t}}, \quad x_{t|T} = x_{t|t} + J_t (x_{t+1|T} - x_{t+1|t}), \quad (93)$$

$$P_{t|T} = P_{t|t} + J_t^2 (P_{t+1|T} - P_{t+1|t}). \quad (94)$$

The recursion is initialized at the final filtered values $x_{T|T}$ and $P_{T|T}$.

The smoothed predictor of the transitory output shock is

$$\widehat{\eta}_{it} = \widetilde{RR}_{it} - x_{t|T}. \quad (95)$$

C.4 Recovering Hicks-neutral productivity

The main IPC estimator recovers the composite

$$(\widehat{\omega_{N,it} + \eta_{it}})$$

from equations (52)–(53). The signal-extraction step yields the firm-year predictor

$$\widehat{\omega}_{N,it}^{SE} = (\widehat{\omega_{N,it} + \eta_{it}}) - \widehat{\eta}_{it}. \quad (96)$$

D Variable Definitions

- **Investment (I_{it}):** The value of current-period investment in equipment goods, excluding buildings, land, and financial assets. Investment made in period t becomes productive in period $t + 1$.
- **Capital (K_{it}):** Real capital constructed by the perpetual inventory recursion and industry-size initialization described in Section 3, using reported real net capital when available and annual depreciation $d_K = 0.15$.
- **Labor (L_{it}):** Total hours worked by all employees of the firm in year t .
- **Materials (M_{it}):** The real value of intermediate consumption, comprising raw materials, energy, and purchased services. Nominal intermediate consumption expenditures are deflated by a firm-specific intermediate consumption price index constructed from price variations reported in the ESEE.
- **Output (Y_{it}):** The real value of goods and services produced, obtained by deflating nominal production values by a firm-specific output price index constructed using the Paasche index formula.
- **Nominal output and the variable-cost ratio:** $P_{it}^Y Y_{it}$ is nominal goods-and-services production, the denominator used in the model's expenditure shares. The symbol R_{it} denotes total labor and materials expenditure divided by nominal output, not nominal sales.
- **Wage ($W_{L,it}$):** The hourly wage rate, defined as total labor compensation divided by total effective hours worked.
- **Output Price Index (P_{it}^Y):** A firm-specific Paasche price index constructed from price variations of output sales reported in the ESEE, following the methodology of Jaumandreu and Lin (2018).
- **Materials Price Index (P_{it}^M):** A firm-specific Paasche price index constructed from price variations of intermediate consumption reported in the ESEE.
- **Labor share ($s_{L,it}$):** The ratio of total labor compensation to total revenue, $s_{L,it} \equiv W_{L,it} L_{it} / (P_{it} Y_{it})$. It is revenue-denominated and therefore differs from the value-added labor share.
- **Materials share ($s_{M,it}$):** The ratio of total materials expenditure to total revenue, $s_{M,it} \equiv W_{M,it} M_{it} / (P_{it} Y_{it})$.

E Digitalization Index

Table 17: Variables used to construct the digitalization index

Dimension	Variable	Definition
Automation	Use of robots	Whether the company uses robots in the production process (Yes/No)
	Use of CNC machine tools	Whether the company uses computer numerically controlled machine tools in the production process (Yes/No)
	Use of computer-aided design	Whether the company makes use of computer-assisted design (Yes/No)
	Use of flexible systems	Whether the company combines machine tools controlled by computer, robotics, and/or CAD through a mainframe computer (Yes/No)
	Use of LAN	Whether the company uses a local area network in its manufacturing activities (Yes/No)
Interactions with stakeholders	Ownership of internet domain	Whether the company has its own internet domain (Yes/No)
	Ownership of a webpage	Whether the company's web page is stored on its own servers (Yes/No/NA)
	Sales to companies via internet	Availability of a sales system for other companies through internet (Yes/No/NA)
	Sales to final consumer via internet	Availability of a sales system for final customers through internet (Yes/No/NA)
	Purchases from suppliers online	Whether goods or services are purchased through the internet (Yes/No/NA)
Digital-related human capital	External expenditure on IT training	Value of external expenditure on employee training in computing and information technologies (Euros)
IT technologies	Investment in computer equipment	Share of purchases of information processing equipment in total tangible fixed asset purchases (%)
	Hiring of software programming	Whether the company contracts out software-programming services (Yes/No)

Notes: The global index contains the 13 variables shown. ESEE adoption codes are recoded as 1 for Yes and 0 for No; non-response codes are treated as missing before aggregation. Computer-equipment investment is converted from a percentage of total capital-goods investment into an absolute flow before capitalization. Computer-equipment investment and IT-training expenditure are capitalized with 15% depreciation. Quadrennial indicators are carried forward within firms for at most three years. Section 3.1 describes scaling, normalization, and the treatment of missing components at aggregation.